

Chapter VII

Julia Sets

1 The Escape Time Algorithm for Computing Pictures of IFS Attractors and Julia Sets

Let us consider the dynamical system $\{\mathbb{R}^2; f\}$, where $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by

$$f(x, y) = \begin{cases} (2x, 2y - 1) & \text{if } y > 0.5, \\ (2x - 1, y) & \text{if } x > 0.5 \text{ and } y \leq 0.5, \\ (2x, 2y) & \text{otherwise.} \end{cases}$$

This dynamical system is related to the IFS $\{\mathbb{R}^2; w_1, w_2, w_3\}$, where

$$\begin{aligned} w_1(x, y) &= (0.5x, 0.5y + 0.5), \\ w_2(x, y) &= (0.5x + 0.5, 0.5y), \\ w_3(x, y) &= (0.5x, 0.5y). \end{aligned}$$

The attractor of the IFS is a Sierpinski triangle Δ with vertices at $(0, 0)$, $(0, 1)$, and $(1, 0)$. The relationship between the dynamical system $\{\mathbb{R}^2; f\}$ and the IFS $\{\mathbb{R}^2; w_1, w_2, w_3\}$ is that $\{\Delta; f\}$ is a shift dynamical system associated with the IFS. (Shift dynamical systems are discussed in Chapter IV, section 4.) One readily verifies that f restricted to Δ satisfies

$$f(x, y) = \begin{cases} w_1^{-1}(x, y) & \text{if } (x, y) \in w_1(\Delta) \setminus \{(0, 0.5), (0.5, 0.5)\}, \\ w_2^{-1}(x, y) & \text{if } (x, y) \in w_2(\Delta) \setminus \{(0.5, 0)\}, \\ w_3^{-1}(x, y) & \text{if } (x, y) \in w_3(\Delta). \end{cases}$$

In particular, f maps Δ onto itself. The dynamical system $\{\mathbb{R}^2; f\}$ is an extension of the shift dynamical system $\{\Delta; f\}$ to $\mathbb{R}^2$. The situation is illustrated in Figure VII.189.

Let d denote the Euclidean metric on $\mathbb{R}^2$. The shift dynamical system $\{\mathbb{R}^2; f\}$ is “expanding”: for any pair of points x_1, x_2 lying in any one of the three domains associated with f , we have

$$d(f(x_1), f(x_2)) = 2d(x_1, x_2),$$

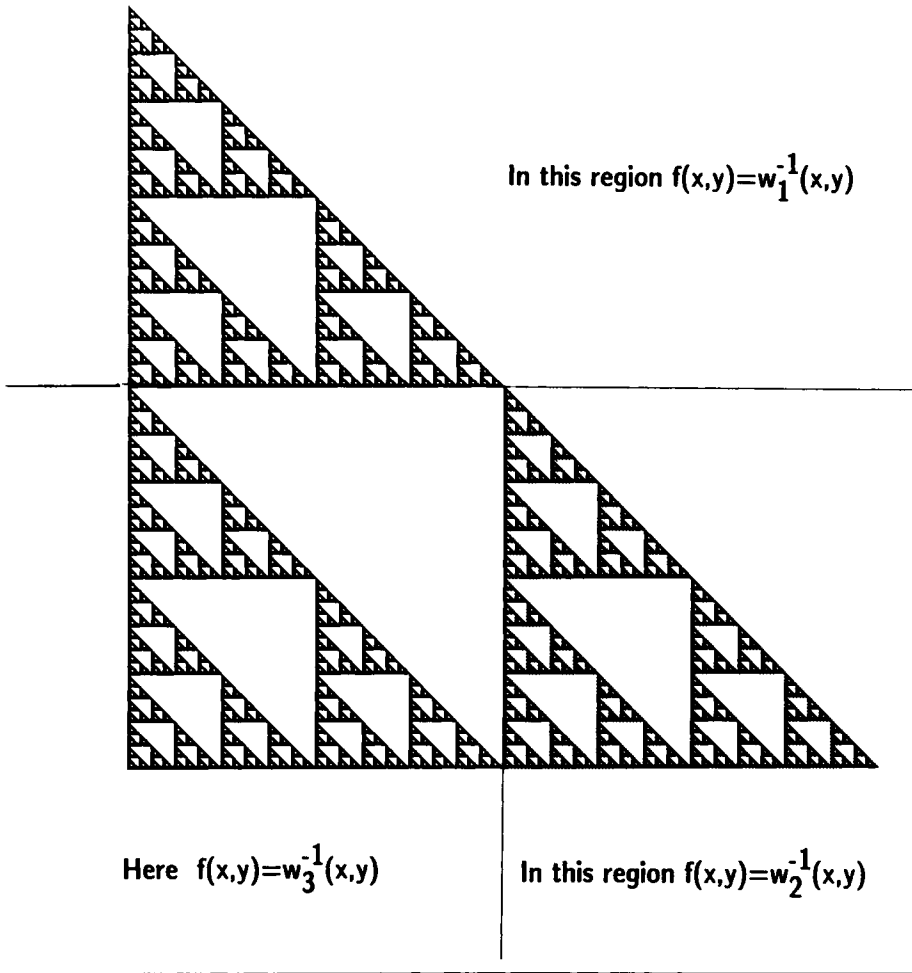


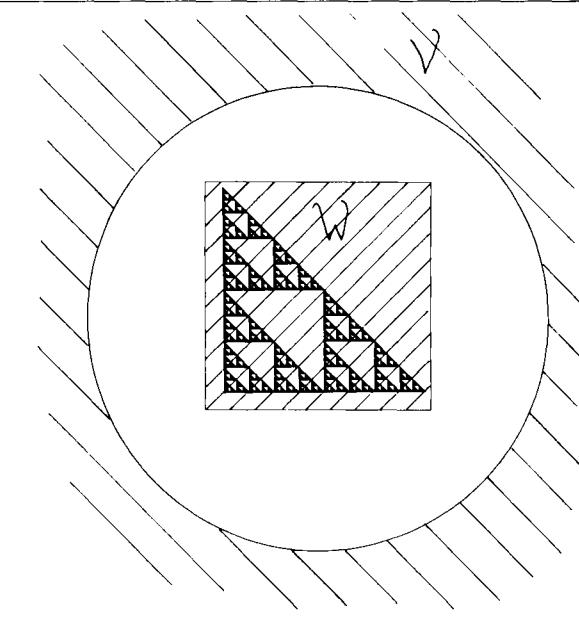
Figure VII.189. The dynamical system $\{\mathbb{R}^2, f\}$ is obtained by extending the definition of a shift dynamical system on a Sierpinski triangle to all of $\mathbb{R}^2$.

One can prove that the orbit $\{f^{on}(x)\}_{n=0}^{\infty}$ diverges toward infinity if x does not belong to Δ . That is

$$d(O, f^{on}(x)) \rightarrow \infty \text{ as } n \rightarrow \infty \text{ for any point } x \in \mathbb{R}^2 \setminus \Delta.$$

What happens if we compute numerically the orbit of a point $x \in \Delta$? Recall that the fractal dimension of Δ is $\log(3)/\log(2)$. This tells us that Δ is “thin” compared to $\mathbb{R}^2$. Hence, although $f(\Delta) = \Delta$, errors in a computed orbit are likely to produce points that do not lie on Δ . This means that, in practice, most numerically computed orbits will diverge, regardless of whether or not the initial point lies on Δ . The Sierpinski triangle Δ is an “unstable” invariant set for the transformation $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. It is a “repulsive” fixed point for the transformation $f : \mathcal{H}(\mathbb{R}^2) \rightarrow \mathcal{H}(\mathbb{R}^2)$. It is an attractive fixed point for the transformation $W : \mathcal{H}(\mathbb{R}^2) \rightarrow \mathcal{H}(\mathbb{R}^2)$, where $W = w_1 \cup w_2 \cup w_3$ is defined in the usual manner.

Figure VII.190. How long do orbits of points in $\mathcal{W}$ take to arrive in $\mathcal{V}$? We expect that the number of iterations required should tell us something about the structure of Δ .



Intuitively, we expect that orbits of the dynamical system $\{\mathbb{R}^2; f\}$ that start close to Δ should “take longer to diverge” than those that start far from Δ . How fast do different orbits diverge? Here we describe a numerical, computergraphical experiment to compare the number of iterations required for the orbits of different points to escape from a ball of large radius, centered at the origin. Let (a, b) and (c, d) , respectively, denote the coordinates of the lower left corner and the upper right corner of a closed, filled rectangle $\mathcal{W} \subset \mathbb{R}^2$. Let M denote a positive integer, and define an array of points in $\mathcal{W}$ by

$$x_{p,q} = \left(a + p \frac{(c-a)}{M}, b + q \frac{(d-b)}{M} \right) \text{ for } p, q = 0, 1, 2, \dots, M.$$

In the experiment these points will be represented by pixels on a computer graphics display device. We compare the orbits $\{f^{on}(x_{p,q}) : \}_{n=0}^{\infty}$ for $p, q = 0, 1, 2, \dots, M$.

Let R be a positive number, sufficiently large that the ball with center at the origin and radius R contains both Δ and $\mathcal{W}$. Define

$$\mathcal{V} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 > R\}.$$

A possible choice for the rectangle $\mathcal{W}$ and the set $\mathcal{V}$, in relation to Δ , is illustrated in Figure VII.190. In order that the comparison of orbits provides information about Δ , one should choose $\mathcal{W}$ so that $\mathcal{W} \cap \Delta \neq \emptyset$.

Let $numits$ denote a positive integer. The following program computes a finite set of points

$$\{f^{o1}(x_{p,q}), f^{o2}(x_{p,q}), f^{o3}(x_{p,q}), \dots, f^{on}(x_{p,q})\}$$

belonging to the orbit of $x_{p,q} \in \mathcal{W}$, for each $p, q = 1, 2, \dots, M$. The total number

of points computed on an orbit is at most *numits*. If the set of computed points of the orbit of $x_{p,q}$ does not include a point in $\mathcal{V}$ when $n = \text{numits}$, then the computation passes to the next value of (p, q) . Otherwise the pixel corresponding to $x_{p,q}$ is rendered in a color indexed by the first integer n such that $f^{on}(x_{p,q}) \in \mathcal{V}$, and then the computation passes to the next value of (p, q) . This provides a computergraphical method for comparing how long the orbits of different points in $\mathcal{W}$ take to reach $\mathcal{V}$.

The program is written in BASIC. It runs without modification on an IBM PC with Enhanced Graphics Adaptor and Turbobasic. On any line the words preceded by a ' are comments: they are not part of the program.

Program 1. ((Example of the Escape Time Algorithm))

```

numits=20: a=0 : b=0: c=1 : d=1 : M=100 'Define viewing
                                         window, W, and numits.

R=200 'Define the region V.

screen 9: cls 'Initialize graphics.

for p=1 to M

for q=1 to M

x = a + (c-a)*p/M : y = b + (d-b)*q/M 'Specify the initial
                                         point of an orbit, x(p,q).

for n=1 to numits 'Compute at most numits points on the orbit
                  of x(p,q).

    if y > 0.5 then 'Evaluate  $f^y$  applied to the
                    previous point on the orbit.

        x = 2*x : y = 2*y - 1

    elseif x > 0.5 then 'THE FORMULA FOR THE
                        FUNCTION f(x)

        x = 2*x - 1 : y = 2*y

    else

        x = 2*x : y = 2*y

    end if

150 if x*x + y*y > R then 'If the most recently computed
                        point lies in  $\mathcal{V}$  then...
```

```

160 pset(p,q),n : n = numits '...render the pixel x(p,q)
                                in color n, and go to the next
    (p,q).

170 end if

if instat then end 'Stop computing if any key is pressed!

next n : next q : next p

end

```

Color Plate 6 shows the result of running a version of Program 1 on a Masscomp 5600 workstation with Aurora graphics.

In Figure VII.191 we show the result of running a version of Program 1, but this time in black and white. A point is plotted in black if the number of iterations required to reach $\mathcal{V}$ is an odd integer, or if the orbit of the point does not reach $\mathcal{V}$ during the first *numits* iterations.

In Figure VII.192 we show the result of running a version of Program 1, with $(a, b) = (0, 0)$, $(c, d) = (5 \times 10^{-18}, 5 \times 10^{-18})$, and *numits* = 65. This viewing window is minute. See also Color Plate 7. Now you should be convinced that Δ is not simplified by magnification.

The dynamical system $\{\mathbb{R}^2; f\}$ contains deep information about the “repelling” set Δ . Some of this information is revealed by means of the Escape Time Algorithm. The orbits of points that lie close to Δ do indeed appear to take longer to escape from $\mathbb{R}^2 \setminus \mathcal{V}$ than those of points which lie further away.

Examples & Exercises

1.1. Let $\{\mathbb{R}^2, f\}$ denote the dynamical system defined at the start of this chapter and let Δ denote the associated Sierpinski triangle. Prove that the orbit $\{f^{on}(x)\}_{n=0}^{\infty}$ diverges, for each $x \in \mathbb{R}^2 \setminus \Delta$. That is, prove that $d(O, f^{on}(x)) \rightarrow \infty$ as $n \rightarrow \infty$ for each $x \in \mathbb{R}^2 \setminus \Delta$.

1.2. Rewrite Program 1 in a form suitable for your own computergraphical environment, then run it and obtain hardcopy of the output.

1.3. If the Escape Time Algorithm is applied to the dynamical system

$$\{\mathbb{R}^2; f(x, y) = (2x, 2y)\},$$

what will be the general appearance of resulting colored regions?

1.4. By changing the window size in Program 1, obtain images of “zooms” on the Sierpinski triangle. For example, use the following windows: $(0, 0) - (0.5, 0.5)$; $(0, 0) - (0.25, 0.25)$; $(0, 0) - (0.125, 0.125)$; ... How must the total number of iterations, *numits*, be adjusted as a function of window size in order

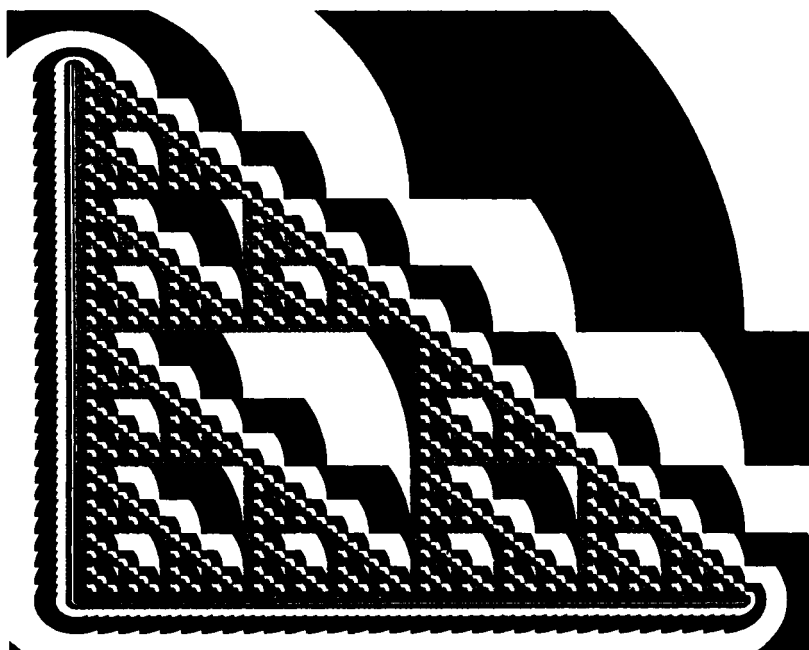


Figure VII.191. Output from a modified version of Program 1. A pixel is rendered in black if either the number of iterations required to reach V is an odd integer, or the orbit does not reach V during the first *numits* iterations.

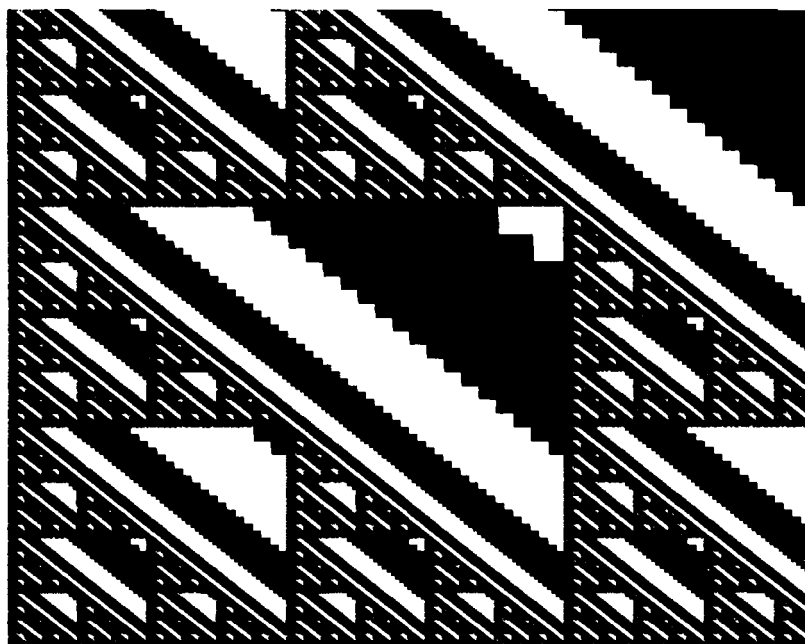


Figure VII.192. Here we show the result of running a version of Program 1, with $(a, b) = (0, 0)$, $(c, d) = (5 \times 10^{-18}, 5 \times 10^{-18})$, and *numits* = 65. This viewing window is minute, yet the computation time was not significantly increased. If we did not know it before, we are now convinced that Δ is not simplified by magnification.

that (approximately) the quality of the images remains uniform? Make a graph of the total number of iterations against the window size. Is there a possible relationship between the behavior of *numits* as a function of window size, and the fractal dimension of the Sierpinski triangle? Make a hypothesis and test it experimentally.

Here we construct another example of a dynamical system whose orbits “try to escape” from the attractor of an IFS. This time we treat an IFS whose attractor has a nonempty interior. Consider the hyperbolic IFS $\{\mathbb{R}^2; w_1, w_2\}$, where

$$\begin{aligned} w_1 \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 & -s^{-1} \\ s^{-1} & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ w_2 \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 & -s^{-1} \\ s^{-1} & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \text{and } s &= \sqrt{2}. \end{aligned}$$

The attractor of this IFS is a closed, filled rectangle, which we denote here by $\blacksquare$. This attractor is the union of two copies of itself, each scaled by a factor $1/\sqrt{2}$, rotated about the origin anticlockwise through 90° , and then translated horizontally, one copy to the left and one to the right. The inverse transformations are

$$\begin{aligned} w_1^{-1} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 & s \\ -s & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ s \end{bmatrix}, \text{ and} \\ w_2^{-1} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 & s \\ -s & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0 \\ s \end{bmatrix}. \end{aligned}$$

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$f(x, y) = \begin{cases} w_1^{-1}(x, y) & \text{when } x > 0 \\ w_2^{-1}(x, y) & \text{when } x \leq 0. \end{cases}$$

Then the dynamical system $\{\mathbb{R}^2; f\}$ is an extension of the shift dynamical system $\{\blacksquare; f\}$ to $\mathbb{R}^2$.

What happens when we apply the Escape Time Algorithm to this dynamical system? To see, one can replace the function $f(x)$ in Program 1 by

```
if x > 0 then
```

```
newx = s*y : newy = -s*x + s
```

```
else
```

```
newx = s*y : newy = -s*x - s
```

```
end if
```

```
x = newx : y = newy
```

```
'THE FORMULA FOR THE  
FUNCTION f(x)
```

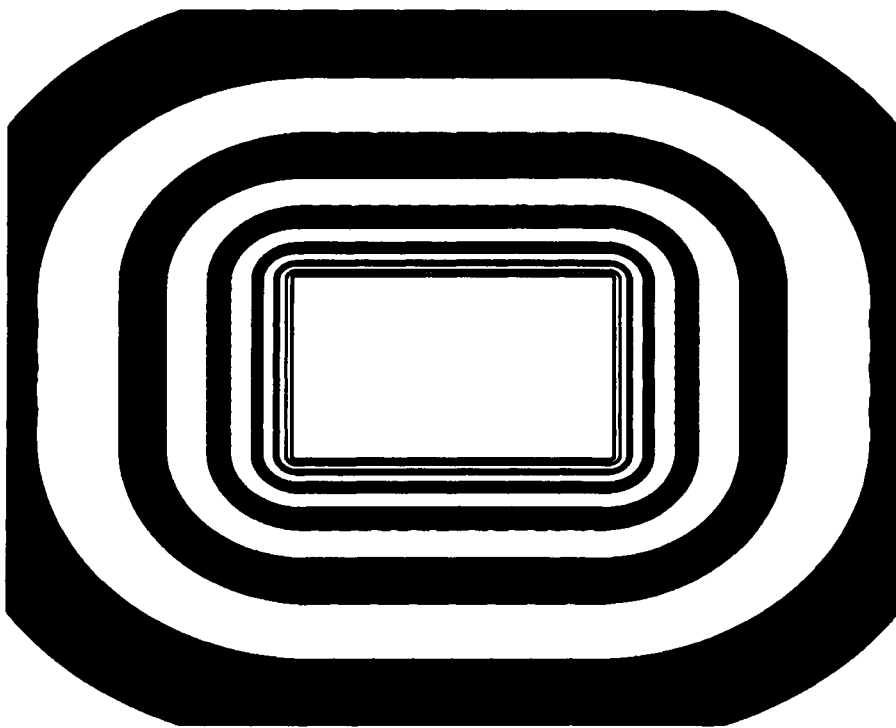


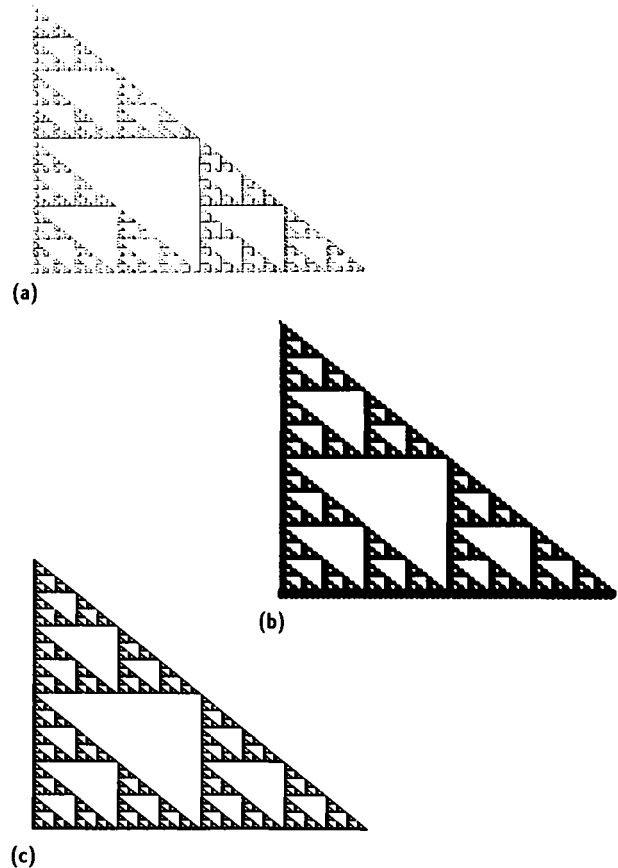
Figure VII.193. An image of an IFS attractor computed using the Escape Time Algorithm. This time the attractor of the IFS is a filled rectangle and the computed orbits of points in ■ seem never to escape.

Results of running Program 1, thus modified, with the window $\mathcal{W}$ and the escape region $\mathcal{V}$ chosen appropriately, are shown in Figure VII.193.

It appears that the orbits of points in the interior ■ do not escape. This is not surprising. The fractal dimension of the attractor of the IFS is the same as the fractal dimension of $\mathbb{R}^2$, so small computational errors are unlikely to knock the orbit off the invariant set. It also appears that the orbits of points that lie in $\mathbb{R}^2 \setminus \blacksquare$ reach $\mathcal{V}$ after fewer and fewer iterations, the farther away from ■ they start.

Again we see that the Escape Time Algorithm provides a means for the computation of the attractor of an IFS. Indeed, we have here the bare bones of a new algorithm for computing images of the attractors of some hyperbolic IFS on $\mathbb{R}^2$. Here are the main steps. (a) Find a dynamical system $\{\mathbb{R}^2; f\}$ which is an extension of a shift dynamical system associated with the IFS, and which tends to transform points off the attractor of the IFS to new points that are farther away from the attractor. (This is always possible if the IFS is totally disconnected. The tricky part is to find a formula for $f(x)$, one which can be input conveniently into a computer. In the case of affine transformations in $\mathbb{R}^2$, one can often define the extensions of the domains of the inverse transformations with the aid of straight lines.) (b) Apply the Escape Time Algorithm, with $\mathcal{V}$ and $\mathcal{W}$ chosen appropriately, but plot only those points whose numerical orbits require sufficiently many iterations before they reach $\mathcal{V}$. For example,

Figure VII.194. Images of an IFS attractor computed using the Escape Time Algorithm. Only points whose orbits have not escaped from $\mathbb{R} \setminus \mathcal{V}$ after *numits* iterations are plotted. The value for *numits* must be chosen not too large, as in (a); and not too small, as in (b); but just right, as in (c).



in Program 1 as it stands, one can replace the three lines 150, 160, and 170 by the two lines

```
150 if n = numits then pset(p,q),1
```

```
160 if x*x + y*y > R then n = numits
```

and define *numits* = 10. If the value of *numits* is too high, then very few points will not escape from $\mathcal{W}$ and a poor image of Δ will result. If the value of *numits* is too low, then a coarse image of the Δ will be produced. An image of an IFS attractor computed using the Escape Time Algorithm, modified as described here, is shown in Figure VII.194.

Color Plates 8–12 show the results of applying the Escape Time Algorithm to the dynamical system associated with various hyperbolic IFS in $\mathbb{R}^2$. In each case the maps are affine, and the shift dynamical system associated with the IFS has been extended to $\mathbb{R}^2$.

Examples & Exercises

1.5. Modify your version of Program 1 to compute images of the attractor of the IFS $\{C; w_1(z) = re^{i\theta}z - 1, w_2(z) = re^{i\theta}z + 1\}$, when $r = 1/\sqrt{2}$ and $\theta = \pi/2$.

1.6. Show that it is possible to define a dynamical system $\{C; f\}$ which extends to C the shift dynamical system associated with the IFS

$$\{C; w_1(z) = re^{i\theta}z - 1, w_2(z) = re^{i\theta}z + 1\},$$

for any $\theta \in [0, 2\pi)$, provided that the positive real number r is chosen sufficiently small. Note that this can be done in such a way that f is continuous.

1.7. Let $\{A; f\}$ denote the shift dynamical system associated with a totally disconnected hyperbolic IFS in $\mathbb{R}^2$. A denotes the attractor of the IFS. Show that there are many ways to define a dynamical system $\{\mathbb{R}^2; g\}$ so that $f(x) = g(x)$ for all $x \in A$.

The Escape Time Algorithm can be applied, often with interesting results, to any dynamical system of the form $\{\mathbb{R}^2; f\}$, $\{C; f\}$, or $\{\hat{C}; f\}$. One needs only to specify a viewing window $\mathcal{W}$ and a region $\mathcal{V}$, to which orbits of points in $\mathcal{W}$ might escape. The result will be a “picture” of $\mathcal{W}$ wherein the pixel corresponding to the point z is colored according to the smallest value of the positive integer n such that $f^{on}(z) \in \mathcal{V}$. A special color, such as black, may be reserved to represent points whose orbits do not reach $\mathcal{V}$ before $(numits + 1)$ iterations.

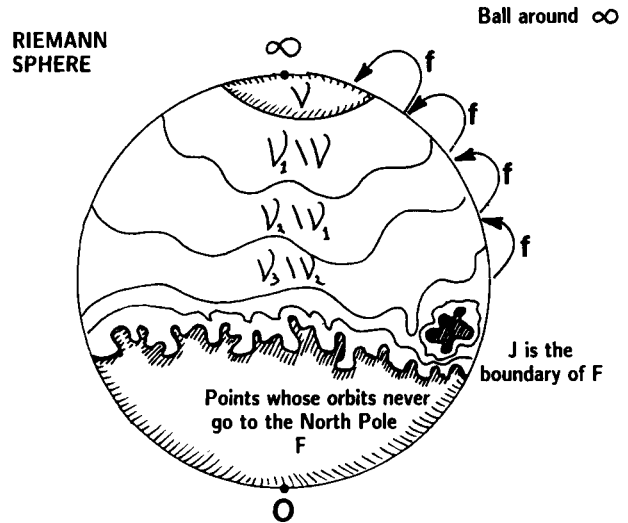
What would happen if the Escape Time Algorithm were applied to the dynamical system $f: \hat{C} \rightarrow \hat{C}$ defined by $f(z) = z^2$? This transformation can be expressed $f(x, y) = (x^2 - y^2, 2xy)$. From the discussion of the quadratic transformation in Chapter III, section 4, we know that the orbits of points in the complement of the unit disk $F = \{z \in C : |z| \leq 1\}$ converge to the point at infinity. Orbits of points in the interior of F converge to the origin. So if $\mathcal{W}$ is a rectangle that contains F and if the radius R , which defines $\mathcal{V}$, is sufficiently large, then we expect that the Escape Time Algorithm would yield pictures of F surrounded by concentric rings of different colors. The reader should verify this!

F is called the *filled Julia set* associated with the polynomial transformation $f(z) = z^2$. The boundary of F is called the *Julia set* of f , and we denote it by J . It consists of the circle of radius 1 centered at the origin. One can think of J on the Riemann Sphere as being represented by the equator on a globe. This Julia set separates those points whose orbits converge to the point at infinity from those whose orbits converge to the origin. Orbits of points on J itself cannot escape, either to infinity or to the origin. In fact $J \in \mathcal{H}(\hat{C})$ and $f(J) = J = f^{-1}(J)$. It is an “unstable” fixed point for the transformation $f: \mathcal{H}(\hat{C}) \rightarrow \mathcal{H}(\hat{C})$.

Definition 1.1 Let $f: \hat{C} \rightarrow \hat{C}$ denote a polynomial of degree greater than 1. Let F_f denote the set of points in C whose orbits do not converge to the point at infinity. That is,

$$F_f = \{z \in C : \{|f^{on}(z)|\}_{n=0}^{\infty} \text{ is bounded} \}.$$

Figure VII.195. Illustration showing what is going on in the proof of Theorem 1.1. This illustrates the increasing sequence of sets $\{\mathcal{V}_n\}$ of the point at infinity. It also shows the decreasing sequence of sets, K_n , the complements of the latter, which converge to the filled Julia set F_f . In general the origin, O , need not belong to F_f .



This set is called the filled Julia set associated with the polynomial f . The boundary of F_f is called the Julia set of the polynomial f , and it is denoted by J_f .

Theorem 1.1 Let $f : \hat{C} \rightarrow \hat{C}$ denote a polynomial of degree greater than 1. Let F_f denote the filled Julia set of f and let J_f denote the Julia set of f . Then F_f and J_f are nonempty compact subsets of C ; that is, $F_f \in \mathcal{H}(C)$ and $J_f \in \mathcal{H}(C)$. Moreover, $f(J_f) = J_f = f^{-1}(J_f)$ and $f(F_f) = F_f = f^{-1}(F_f)$. The set $\mathcal{V}_\infty = \hat{C} \setminus F_f$ is pathwise-connected.

Proof We outline the proof for the one-parameter family of transformations $f_\lambda : \hat{C} \rightarrow \hat{C}$ defined by

$$f_\lambda(z) = z^2 - \lambda, \quad \text{where } \lambda \in C \text{ is the parameter.}$$

The general case is treated in [Blanchard 1984], [Brolin], [Fatou 1919–20], and [Julia 1918], for example. This outline proof is constructed to provide information about the relationship between the theorem and the Escape Time Algorithm. Some of the ideas and notation used here are illustrated in Figure VII.195.

Let J_λ denote the Julia set for f_λ and let F_λ denote the filled Julia set for f_λ . Let d denote the Euclidean metric on C and let

$$R > 0.5 + \sqrt{0.25 + |\lambda|}.$$

Then it is readily verified that

$$d(O, f(z)) > d(O, z) \quad \text{for all } z \text{ such that } d(O, z) \geq R.$$

Define

$$\mathcal{V} = \{z \in C : |z| > R\} \cup \{\infty\}.$$

Then it follows that

$$f(\mathcal{V}) \subset \mathcal{V}.$$

One can prove that the orbit $\{f^{on}(z)\}$ converges to ∞ for all $z \in \mathcal{V}$. No bounded orbit intersects $\mathcal{V}$. It follows that

$$F_\lambda = \{z \in \hat{C} : f^{on}(z) \notin \mathcal{V} \text{ for each finite positive integer } n\}.$$

That is, F_λ is the same as the set of points whose orbits do not intersect $\mathcal{V}$.

Now consider the sequence of sets

$$\mathcal{V}_n = f^{\circ -n}(\mathcal{V}) \quad \text{for } n = 0, 1, 2, \dots$$

For each nonnegative integer n , $\mathcal{V}_n$ is an open connected subset of $(\hat{C}, \text{Spherical})$. $\mathcal{V}_n$ is open because $\mathcal{V}$ is open and f is continuous. $\mathcal{V}_n$ is connected because of the geometry of the quadratic transformation, described in Chapter III, section 4: The inverse image of a path that joins the point at infinity to any other point on the sphere is a path that contains the point at infinity.

Since $f(\mathcal{V}) \subset \mathcal{V}$ it follows that $\mathcal{V} \subset f^{-1}(\mathcal{V})$. This implies that

$$\mathcal{V} = \mathcal{V}_0 \subset \mathcal{V}_1 \subset \mathcal{V}_2 \subset \mathcal{V}_3 \subset \dots \subset \mathcal{V}_n \subset \dots \quad (1)$$

For each nonnegative integer n ,

$$\mathcal{V}_n = \left\{ z \in \hat{C} : \{z, f^{\circ 1}(z), f^{\circ 2}(z), f^{\circ 3}(z), \dots, f^{\circ n}(z)\} \cap \mathcal{V} \neq \emptyset \right\}.$$

That is, $\mathcal{V}_n$ is the set of points whose orbits require at most n iterations to reach $\mathcal{V}$. Let

$$K_n = \hat{C} \setminus \mathcal{V}_n \quad \text{for } n = 0, 1, 2, 3, \dots$$

Then K_n is the set of points whose orbits do not intersect $\mathcal{V}$ during the first n iterations. That is,

$$K_n = \left\{ z \in \hat{C} : \{z, f^{\circ 1}(z), f^{\circ 2}(z), f^{\circ 3}(z), \dots, f^{\circ n}(z)\} \cap \mathcal{V} = \emptyset \right\}.$$

For each nonnegative integer n , K_n is a nonempty compact subset of the metric space $(\hat{C}, \text{Spherical})$. How do we know that K_n is nonempty? Because we can calculate that f possesses a fixed point $z_f \in C$, by solving the equation

$$f(z_f) = z_f^2 - \lambda = z_f.$$

The orbit of z_f converges to z_f . Hence it cannot belong to $\mathcal{V}_n$ for any nonnegative integer n . Hence $z_f \in K_n$ for each nonnegative integer n .

Equation 1 implies that

$$K_0 \supset K_1 \supset K_2 \supset K_3 \supset \dots \supset K_n \supset \dots$$

It follows that $\{K_n\}$ is a Cauchy sequence in $\mathcal{H}(\hat{C})$. It follows that $\{K_n\}$ converges to

a point in $\mathcal{H}(\hat{C})$. The limit is the set of points whose orbits do not intersect $\mathcal{V}$. Hence

$$F_f = \lim_{n \rightarrow \infty} K_n = \bigcap_{n=0}^{\infty} K_n,$$

and we deduce that F_f belongs to $\mathcal{H}(\hat{C})$.

The equation

$$K_{n+1} = f^{\circ -1}(K_n) \quad \text{for } n = 0, 1, 2, \dots$$

now implies, as in the proof of Theorem 4.1, that

$$F_\lambda = f^{\circ -1}(F_\lambda).$$

Applying f to both sides of this equation, we obtain

$$f(F_\lambda) = F_\lambda.$$

Let us now consider the boundary of F_λ , namely the Julia set J_λ for the dynamical system $\{\hat{C}; f_\lambda\}$. Let $z \in \text{interior}(F_\lambda)$. Then the continuity of f implies $f^{-1}(z) \subset \text{interior}(F_\lambda)$. Hence $F_\lambda \supset f^{-1}(\partial F_\lambda) \supset \partial F_\lambda$. Now suppose that $z \in f^{-1}(\partial F_\lambda)$. Let $\mathcal{O}$ be any open ball that contains z . Since f is analytic, $f(\mathcal{O})$ is an open ball, and it contains $f(z) \in \partial F_\lambda$. Hence $f(\mathcal{O})$ contains a point whose orbit converges to the point at infinity. It follows that $\mathcal{O}$ contains a point whose orbit converges to the point at infinity. Thus $f^{-1}(\partial F_\lambda) \subset \partial F_\lambda$. We conclude that $f^{-1}(\partial F_\lambda) = \partial F_\lambda$ and in particular that $f(\partial F_\lambda) = \partial F_\lambda$. This completes the proof of the theorem.

We summarize some of what we discovered in the course of this proof. The filled Julia set F_λ is the limit of a decreasing sequence of compact sets. Its complement, which we denote by $\mathcal{V}_\infty$, is the limit of an increasing sequence $\{\mathcal{V}_n\}$ of open pathwise-connected sets in $(\hat{C}, \text{spherical})$. That is,

$$\mathcal{V}_\infty = \lim_{n \rightarrow \infty} \mathcal{V}_n = \bigcup_{n=0}^{\infty} \mathcal{V}_n.$$

The latter is called the *basin of attraction* of the point at infinity under the polynomial transformation f_λ . It is connected because each of the sets $\mathcal{V}_n$ is connected. We have

$$\hat{C} = F_\lambda \cup \mathcal{V}_\infty.$$

$\mathcal{V}_\infty$ is open, connected, and nonempty. F_λ is compact and nonempty.

The Escape Time Algorithm provides us with a means for “seeing” the filled Julia sets F_λ , as well as the sequences of sets $\{\mathcal{V}_n\}$ and $\{K_n\}$ referred to in the theorem. Let us look at what happens in the case $\lambda = 1.1$. Define $\mathcal{V}$ by choosing $R = 4$, and let $\mathcal{W} = \{(x, y) : -2 \leq x \leq 2, -2 \leq y \leq 2\}$. The function $f_{\lambda=1.1} : C \rightarrow C$ is given by the formula

$$f_{\lambda=1.1}(x, y) = (x^2 - y^2 - 1.1, 2xy) \quad \text{for all } (x, y) \in C.$$

An example of the result of running the Escape Time Algorithm, with $\mathcal{V}, \mathcal{W}$ and $f : \hat{C} \rightarrow \hat{C}$ thus defined, is shown in Figure VII.196. The black object represents

the filled Julia set $F_{\lambda=1.1}$. The contours separate the regions $\mathcal{V}_{n+1} \setminus \mathcal{V}_n$, for some successive values of n . These contours also represent the boundaries of the regions K_n referred to in the proof of the theorem. We refer to them as escape time contours. Points in $\mathcal{V}_{n+1} \setminus \mathcal{V}_n$ have orbits that reach $\mathcal{V}$ in exactly $(n+1)$ iterations. In Color Plate 13 we show another example of running the Escape Time Algorithm to produce an image of the same set. The regions $\mathcal{V}_{n+1} \setminus \mathcal{V}_n$ are represented by different colors.

Figure VII.197 shows a zoom on an interesting piece of $F_{\lambda=1.1}$, including parts of some escape time contours. This image was computed by choosing $\mathcal{W}$ to be a small rectangular subset of the window used in Figure VII.196.

Figures VII.198(a)–(e) shows pictures of the filled Julia sets F_λ for a set of real values of λ . These pictures also include a number of the escape time contours, to help indicate the location of F_λ . F_0 is a filled disk. As λ increases, the set becomes more and more pinched together until, when $\lambda = 2$, it is the closed interval $[-2, 2]$. For some values of $\lambda \in [0, 2]$, it appears that F_λ has no interior, and is “tree-like”; for other values it seems to possess a roomy interior. It also appears that F_λ is connected for all $\lambda \in [0, 2]$, and totally disconnected when $\lambda > 2$. In the latter case F_λ may be described as a “Cantor-like” set, or as a “dust.” The transition between the totally disconnected set and the connected, bubbly set as the parameter λ is varied reminds us of the transition between the the Cantor set and the Sierpinski triangle, discussed in connection with Figure IV.118.

Examples & Exercises

1.8. Modify your version of the Escape Time Algorithm to allow you to compute pictures of filled Julia sets for the family of quadratic polynomials $z^2 - \lambda$ for

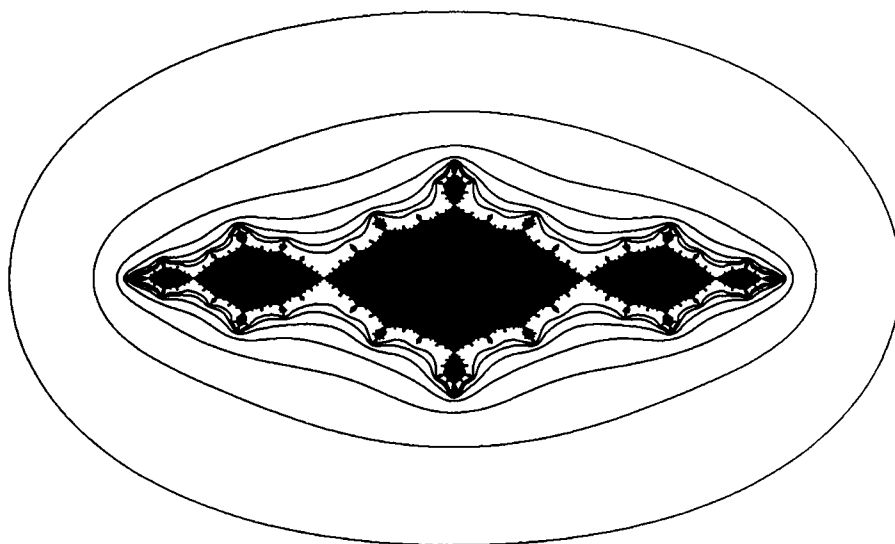


Figure VII.196. The Escape Time Algorithm provides us with a means for “seeing” the filled Julia sets F_λ , as well as the sequences of sets $\{\mathcal{V}_n\}$ and $\{K_n\}$ referred to in Theorem 1. In this illustration, $\lambda = 1.1$. The black object represents the filled Julia set $F_{\lambda=1.1}$. The contours separate the regions $\mathcal{V}_{n-1} \setminus \mathcal{V}_n$, for some successive values of n . These contours also represent the boundaries of the regions K_n referred to in the proof of the theorem.

Figure VII.197. Zoom in on an interesting piece of Figure VII.196.



complex values of λ . Compute a picture of the filled Julia set for $\lambda = i$ and obtain hardcopy of the output.

1.9. Give the iteration formulas, and find a suitable value for R in terms of $|\lambda|$, so that the Escape Time Algorithm can be applied to the complex polynomial $z^3 - \lambda$.

1.10. Study web diagrams associated with $x^2 - \lambda$, for increasing values of $\lambda \in [0, 3]$. Speculate on the relation of these diagrams to the corresponding filled Julia sets.

1.11. Let $\lambda \in [0, 0.7] \cup [0.8, 1.2]$. Let $\mathcal{V}$ to be an open ball of radius 0.00001 centered at the origin. Run the Escape Time Algorithm to the dynamical system $\{C, z^2 - \lambda\}$ with this choice of $\mathcal{V}$. Obtain computergraphical data in support of the hypothesis that, in this case, the algorithm yields approximate pictures of pieces of the closure of $C \setminus F_\lambda$. Design an escape region $\mathcal{V}$ so that, for $\lambda \in [0, 0.7] \cup [0.8, 1.2]$, the Escape Time Algorithm yields approximate pictures of J_λ .

1.12. The Escape Time Algorithm introduces numerical errors in the computation of orbits. These errors should lead to inaccuracies in the computed pictures of Julia sets and IFS attractors. Consider the application to the filled Julia set for $z^2 - 1$. By means of computergraphical experiments, determine the importance of these errors in the images you compute. One way to proceed is to choose successively smaller

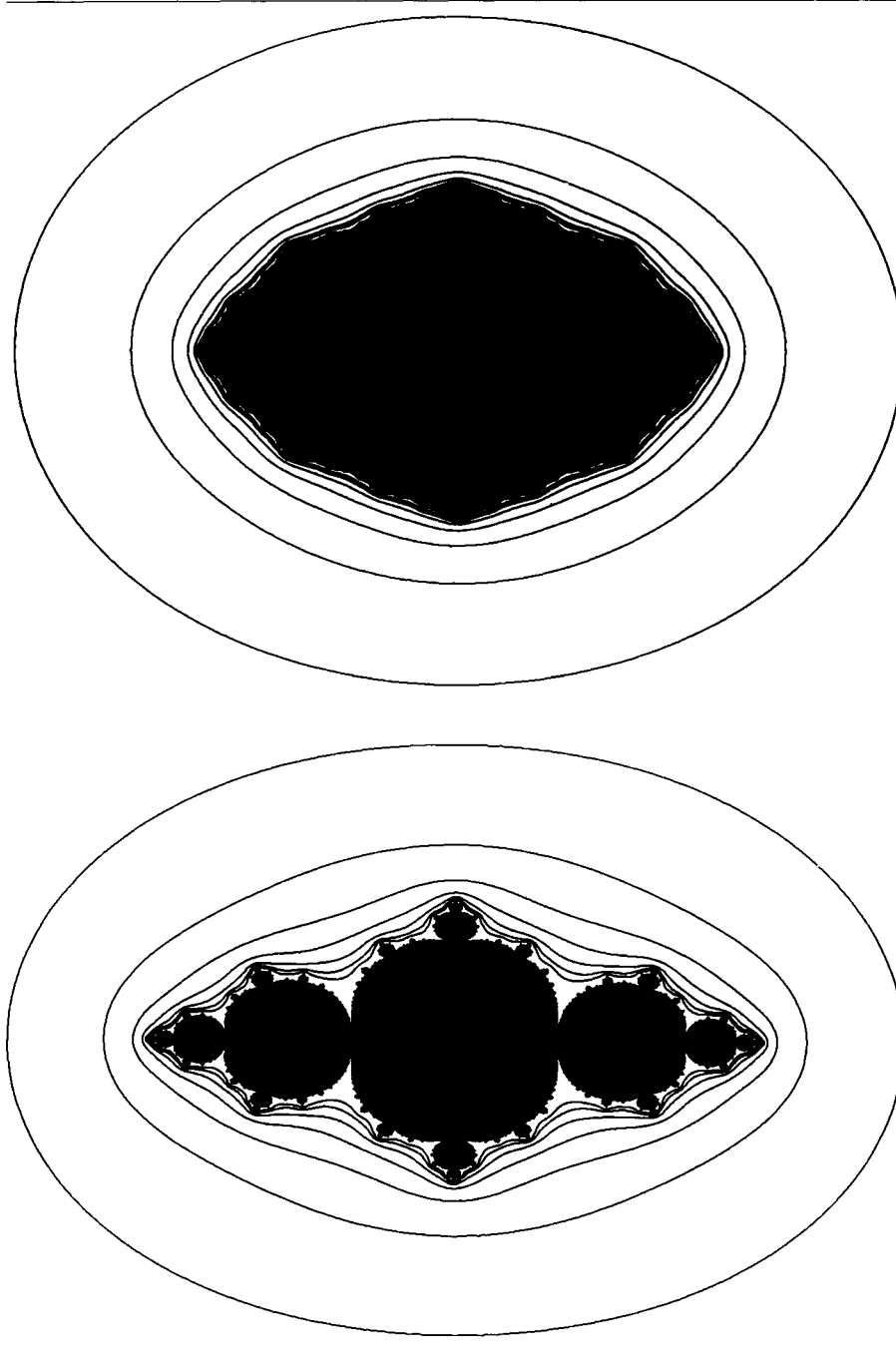


Figure VII.198. (a)–(e) A sequence of Julia set images, as in Figure VII.196, for an increasing sequence of values λ in the range 0 to 3. In (d) and (e) the filled Julia set is the same as the Julia set: the filled Julia set has no interior, so it equals its boundary. In (d) the Julia set is “tree-like.” In (e) the Julia set is totally disconnected.

Figure VII.198. (b)

Figure VII.198. (c)

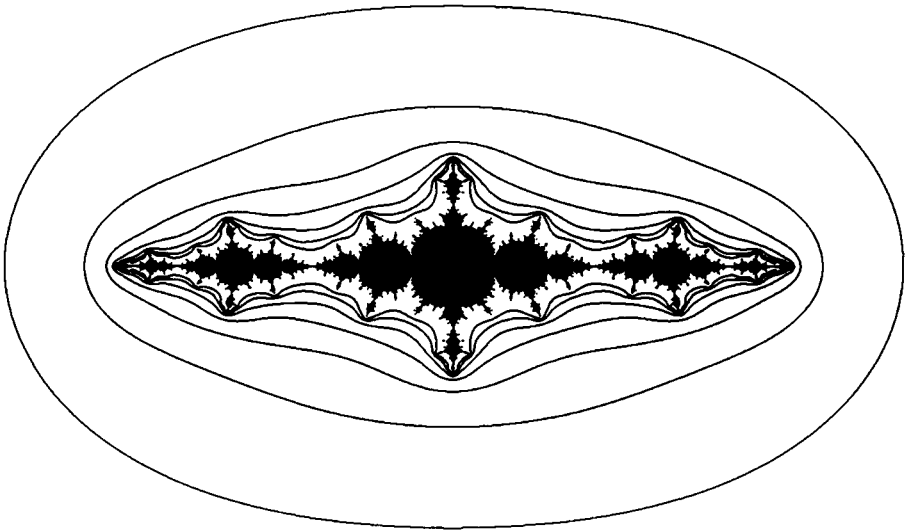


Figure VII.198. (d)

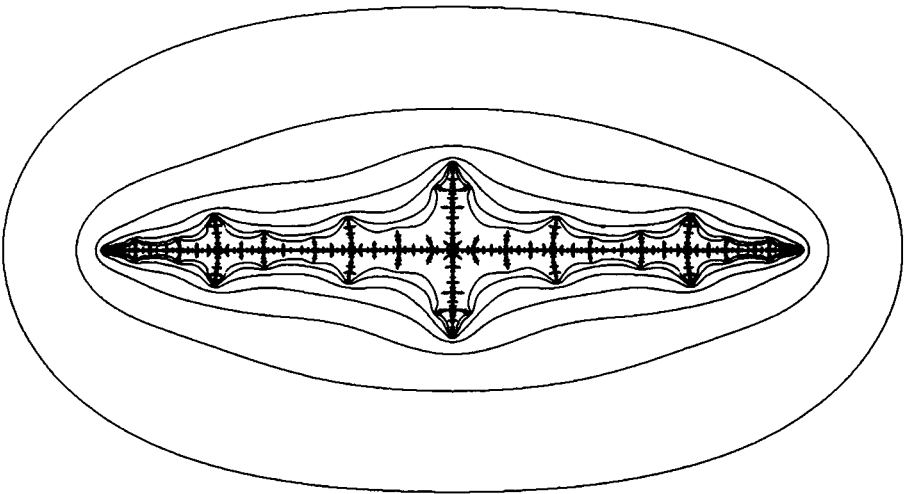
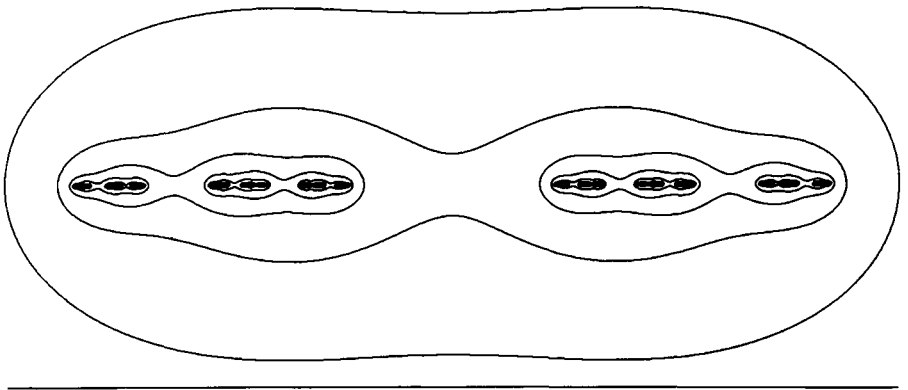


Figure VII.198. (e)



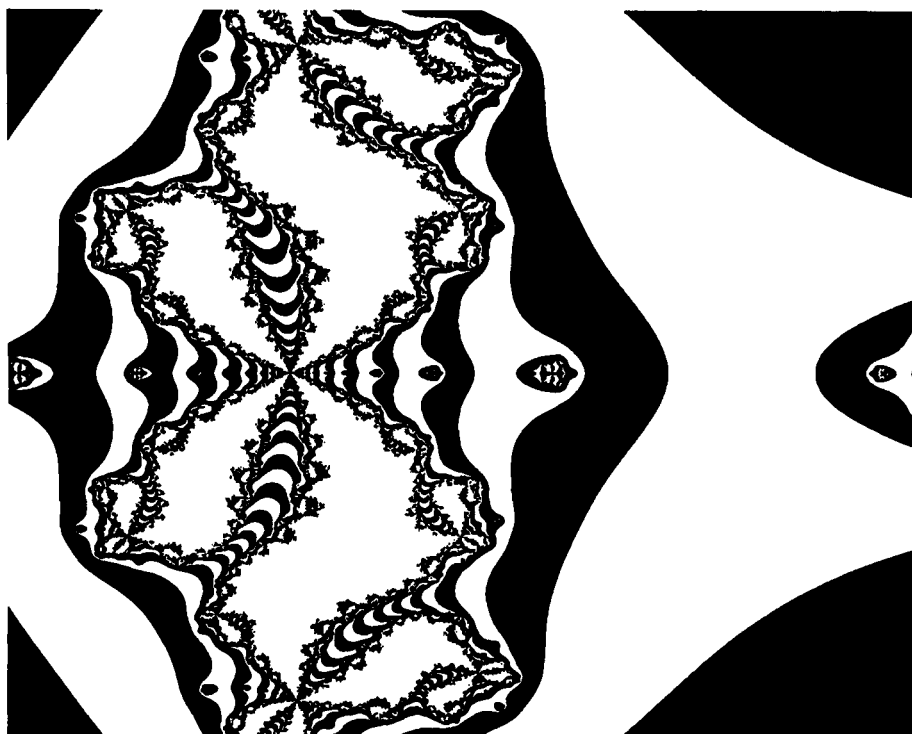


Figure VII.199. This image was computed by applying the Escape Time Algorithm to the dynamical system $\{C; f(z) = z^4 - z - 0.78\}$. The viewing window is $\mathcal{W} = \{(x, y) : -1 \leq x \leq 1, -1 \leq y \leq 1\}$. Can you determine the escape region $\mathcal{V}$?

windows $\mathcal{W}$, which intersect the apparent boundary of the filled Julia set, and to seek the window size at which the quality of computed images seems to deteriorate. (You will need to increase the maximum number of iterations, M , as you zoom.) Can you give evidence to show that the apparently deteriorated images are not, in fact, correct?

1.13. Figure VII.199 was computed by applying the Escape Time Algorithm to the dynamical system $\{C; f(z) = z^4 - z - 0.78\}$. The viewing window is $\mathcal{W} = \{(x, y) : -1 \leq x \leq 1, -1 \leq y \leq 1\}$. Determine the escape region $\mathcal{V}$. Also, you might like to try magnifying one of the little faces in this image.

1.14. The images in Figure VII.200 (a),(b),(c), and (d) represent the nontrivially distinct attractors of all IFS of the form

$$\{\blacksquare; w_1, w_2, w_3\},$$

where the maps are similitudes of scaling factor one-half, and rotation angles in the set $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. The three translations $(0, 0)$, $(1, 0)$, and $(0, 1)$ are used. These IFS are all just-touching. For $i \neq j$ the set $w_i(A) \cap w_j(A)$ is contained in one of the two straight lines $x = 1$ or $y = 1$. Show that, as a result, it is easy to compute these images using the Escape Time Algorithm.

Here are some observations about this “group” of images. Many of them contain

Figure VII.200. (a)–(d) The images in (a), (b), (c), and (d) represent the nontrivially distinct attractors of all IFS of the form $\{\blacksquare; w_1, w_2, w_3\}$, where the maps are similitudes of scaling factor one-half, and the rotation angles are in the set $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. The three translations $(0, 0)$, $(1, 0)$, and $(0, 1)$ are used. These IFS are all just-touching. For $i \neq j$ the set $w_i(A) \cap w_j(A)$ is contained in one of the two straight lines $x = 1$ or $y = 1$. Hence it is easy to compute images of these attractors using the Escape Time Algorithm.

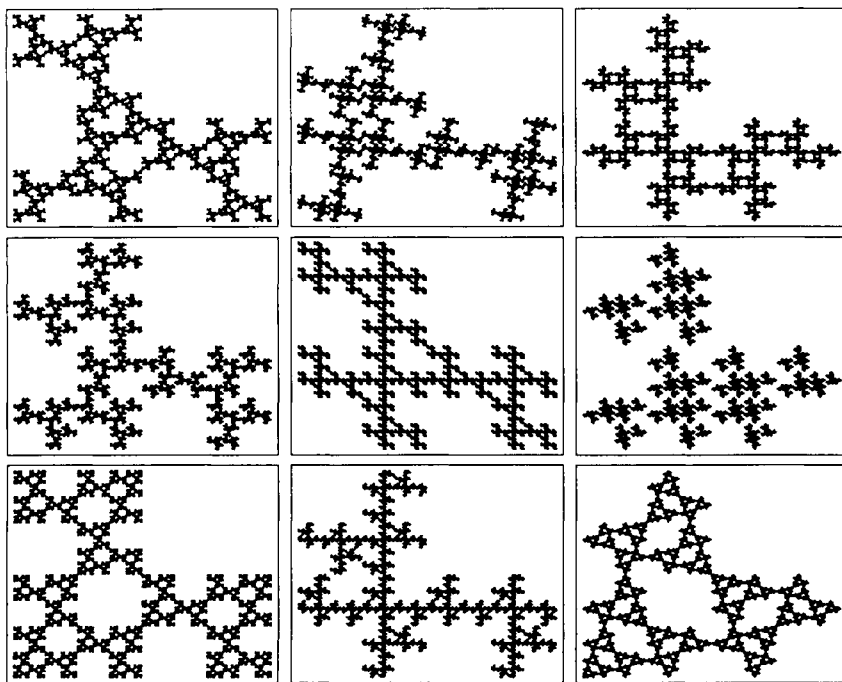
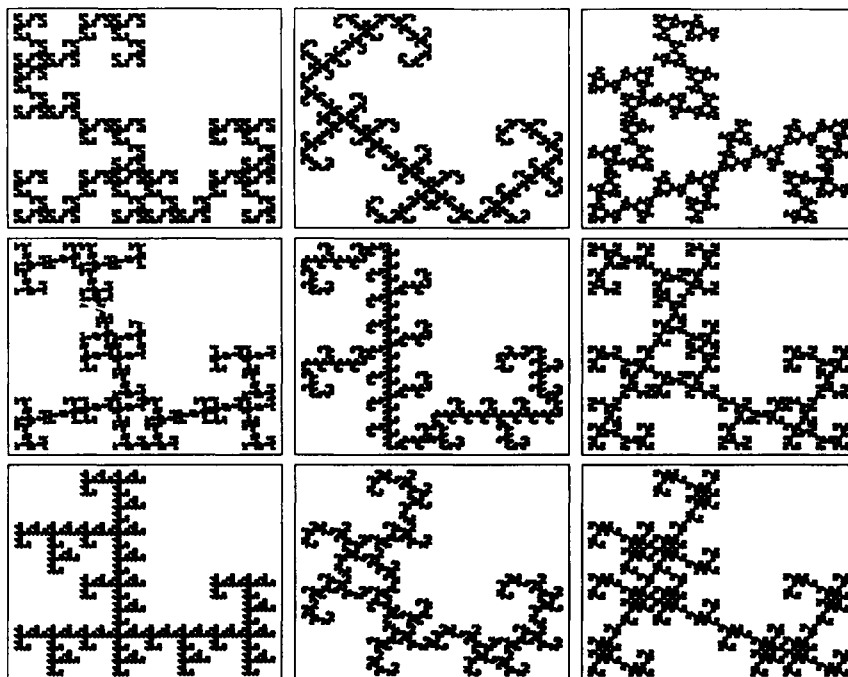


Figure VII.200. (b)



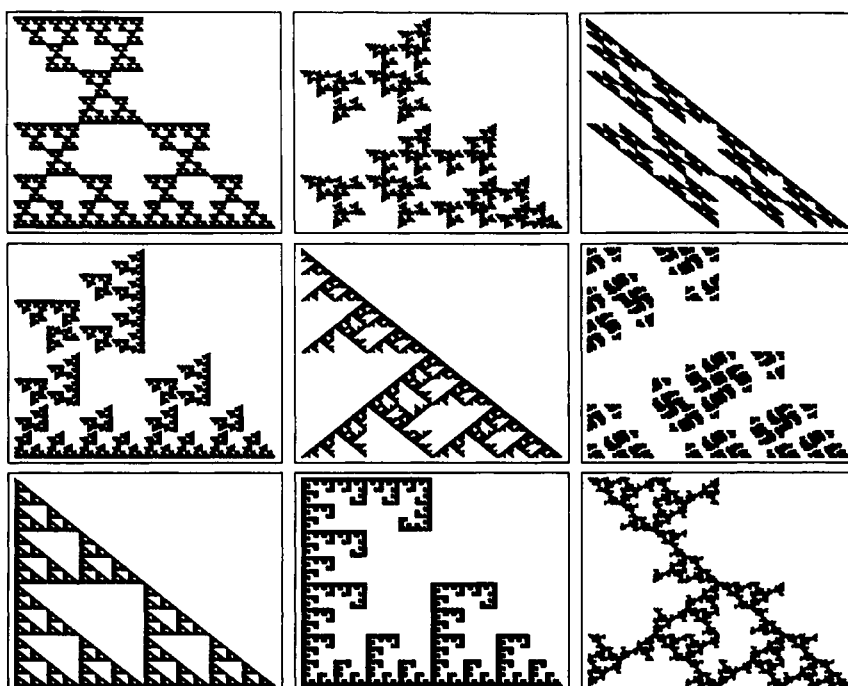


Figure VII.200. (c)

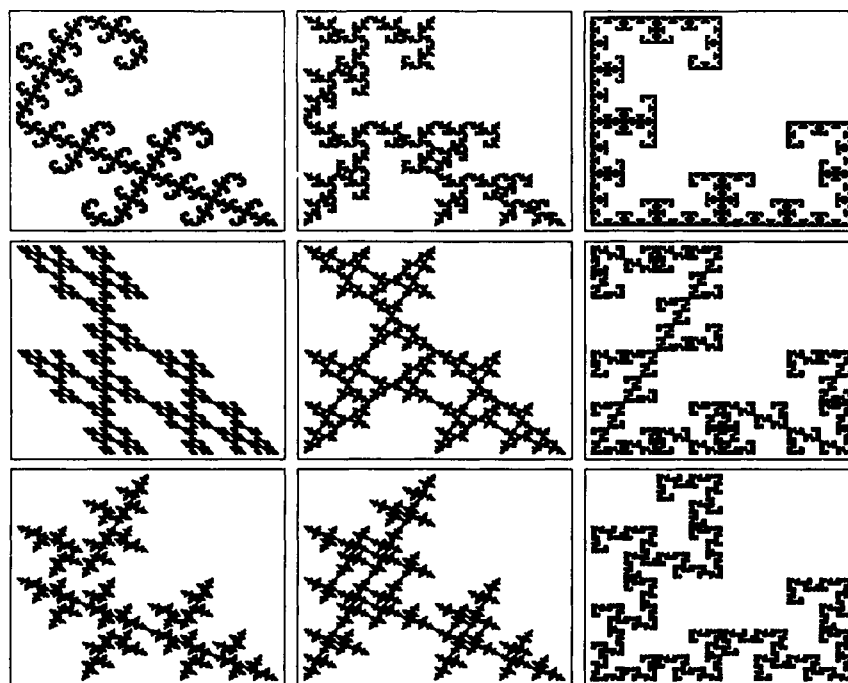


Figure VII.200. (d)

straight lines. They all have the same fractal dimension. They all use approximately the same amount of ink. Many of them are connected. Make some more observations. Can you formalize and prove some of these observations?

1.15. Verify computationally that a “snowflake” curve is a basin boundary for the dynamical system $\{\mathbb{R}^2; f\}$, where for all $(x, y) \in \mathbb{R}^2$,

$$f(x, y) = (0, -1) \text{ if } y < 0;$$

$$f(x, y) = (3x, 3y) \text{ if } y \geq 0 \text{ and } x < -y/\sqrt{3} + 1;$$

$$f(x, y) = ((9 - 3x - 3\sqrt{3}y)/2, (3\sqrt{3} - 3\sqrt{3}x + 3)/2)$$

$$\text{if } y \geq 0 \text{ and } -y/\sqrt{3} + 1 \leq x < 3/2;$$

$$f(x, y) = ((3x - 3\sqrt{3}y)/2, (3\sqrt{3}x + 3y - 6\sqrt{3})/2)/2,$$

$$\text{if } y \geq 0 \text{ and } 3/2 \leq x < y/\sqrt{3} + 2;$$

$$f(x, y) = (9 - 3x, 3y), \text{ if } y \geq 0, \text{ and } x \geq y/\sqrt{3} + 2.$$

2 Iterated Function Systems Whose Attractors Are Julia Sets

In section 1 we learned how to define some IFS attractors and filled Julia sets with the aid of the Escape Time Algorithm applied to certain dynamical systems. In this section we explain how the Julia set of a quadratic transformation can be viewed as the attractor of a suitably defined IFS.

The Escape Time Algorithm compares how fast different points in $\mathcal{W}$ escape to $\mathcal{V}$, under the action of a dynamical system. Which set repels the orbits? From where do the escaping orbits originate? In the case of the dynamical systems considered at the start of section 1, orbits were “escaping from” the attractor of the IFS.

Let $\lambda \in \mathbb{C}$ be fixed. Which set repels the orbits, in the case of the dynamical system $\{\hat{\mathbb{C}}; f_\lambda(z) = z^2 - \lambda\}$? To find out let us consider the inverse of $f_\lambda(z)$. This is provided by a pair of functions, $f^{-1}(z) = \{+\sqrt{z + \lambda}, -\sqrt{z + \lambda}\}$, where, for example, the positive square root of a complex number is that complex root that lies on the nonnegative real axis or in the upper half plane. Explicitly, $\sqrt{z} = \sqrt{x_1 + ix_2} = (a(x_1, x_2), b(x_1, x_2))$ with

$$a(x_1, x_2) = \sqrt{\sqrt{\frac{x_1^2 + x_2^2 + x_1}{2}}} \quad \text{when } x_2 \geq 0,$$

$$a(x_1, x_2) = -\sqrt{\sqrt{\frac{x_1^2 + x_2^2 + x_1}{2}}} \quad \text{when } x_2 < 0,$$

$$b(x_1, x_2) = \sqrt{\sqrt{\frac{x_1^2 + x_2^2 - x_1}{2}}}.$$

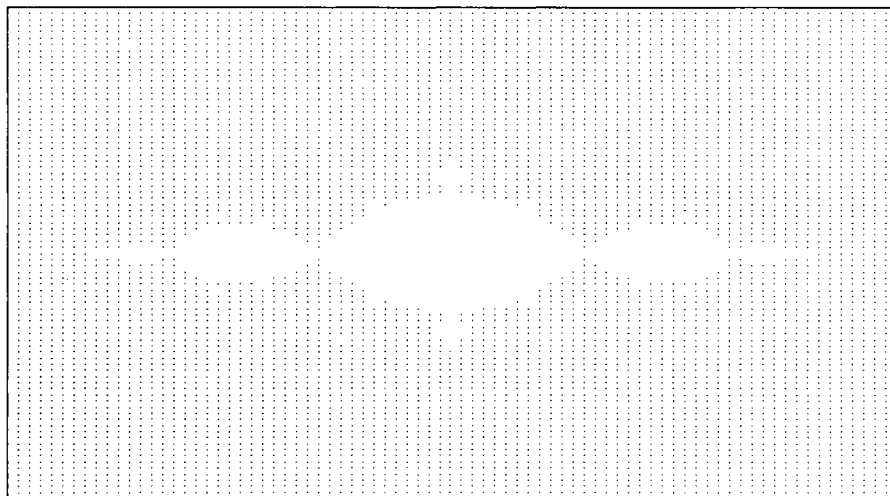


Figure VII.201. (a) and (b) The attractor of the IFS $\{\hat{\mathbb{C}}; w_1(z) = \sqrt{z+1}, w_2(z) = -\sqrt{z+1}\}$ is the Julia set for the transformation $f(z) = z^2 - 1$. (a) illustrates points whose orbits “escape” when the Escape Time Algorithm is applied. (b) shows the results of applying the Random Iteration Algorithm to the IFS, superimposed on (a).

To find the “repelling” set, we must try to run the dynamical system backwards. This leads us to study the IFS

$$\{\hat{\mathbb{C}}; w_1(z) = \sqrt{z+\lambda}, w_2(z) = -\sqrt{z+\lambda}\}.$$

The natural idea is that this IFS has an attractor. This attractor is the set from which points try to flee, under the action of the dynamical system $\{\hat{\mathbb{C}}; z^2 - \lambda\}$.

A few computergraphical experiments quickly suggest a wonderful idea: they suggest that the IFS indeed possesses an attractor, namely the Julia set $J_\lambda = \partial F_\lambda$ for $f_\lambda(z)$. Consider, for example, the case $\lambda = 1$. Figure VII.201 (a) illustrates points in the window $\mathcal{W} = \{z = (x, y) \in \mathbb{C} : -2 \leq x \leq 2, -2 \leq y \leq 2\}$ whose orbits diverge. It was computed using the Escape Time Algorithm. Figure VII.201 (b) shows the results of applying the Random Iteration Algorithm to the above IFS, with $\lambda = 1$ and the same screen coordinates, superimposed on (a). The boundary of the region $F_{\lambda=1}$ is outlined by points on the attractor of the IFS.

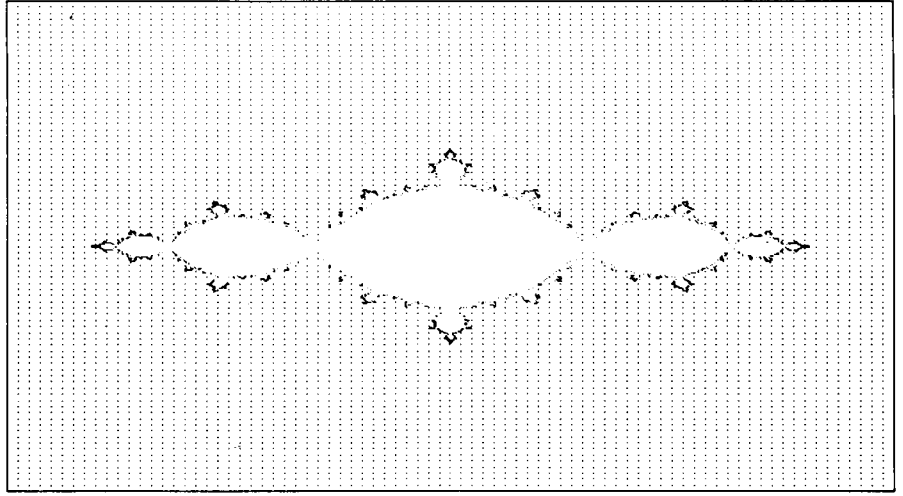
Figures VII.202(a)–(d) show the results of applying the Random Iteration Algorithm to the IFS $\{\hat{\mathbb{C}}; w_1(z) = \sqrt{z+\lambda}, w_2(z) = -\sqrt{z+\lambda}\}$ for various $\lambda \in [0, 3]$. In all cases it appears that the IFS possesses an attractor, and this attractor is the Julia set J_λ .

Perhaps

$$\{\hat{\mathbb{C}}; w_1(z) = \sqrt{z+\lambda}, w_2(z) = -\sqrt{z+\lambda}\}$$

is a hyperbolic IFS with J_λ as its attractor? No, it is not, because $\hat{\mathbb{C}} = w_1(\hat{\mathbb{C}}) \cup w_2(\hat{\mathbb{C}})$. The IFS is not associated with a unique fixed point in the space $\mathcal{H}(\hat{\mathbb{C}})$. In order to make the IFS have a unique attractor, we need to remove some pieces from $\hat{\mathbb{C}}$, to produce a smaller space on which the IFS acts.

Figure VII.201. (b)



Theorem 2.1 Let $\lambda \in \mathbb{C}$. Suppose that the dynamical system $\{\hat{\mathbb{C}}; f(z) = z^2 - \lambda\}$ possesses an attractive cycle $\{z_1, z_2, z_3, \dots, z_p\} \subset \mathbb{C}$. Let ϵ be a very small positive number. Let $\mathbf{X}$ denote the Riemann Sphere $\hat{\mathbb{C}}$ with $(p + 1)$ open balls of radius ϵ removed. (The radius is measured using the spherical metric.) One ball is centered at each point of the cycle, and one ball is centered at the point at infinity, as illustrated in Figure VII.203. Define an IFS by

$$\{\mathbf{X}; w_1(z) = \sqrt{z + \lambda}, w_2(z) = -\sqrt{z + \lambda}\}.$$

Then the transformation W on $\mathcal{H}(\mathbf{X})$, defined by

$$W(B) = w_1(B) \cup w_2(B) \text{ for all } B \in \mathcal{H}(\mathbf{X}),$$

maps $\mathcal{H}(\mathbf{X})$ into itself, continuously with respect to the Hausdorff metric on $\mathcal{H}(\mathbf{X})$. Moreover $W : \mathcal{H}(\mathbf{X}) \rightarrow \mathcal{H}(\mathbf{X})$ possesses a unique fixed point, J_λ , the Julia set for $z^2 - \lambda$. Also

$$\lim_{n \rightarrow \infty} W^{on}(B) = J_\lambda \text{ for all } B \in \mathcal{H}(\mathbf{X}).$$

These conclusions also hold if the orbit of the origin, $\{f^{on}(O)\}$, converges to the point at infinity, and $\mathbf{X} = \hat{\mathbb{C}} \setminus B(\infty, \epsilon)$.

Sketch of proof The fact that W takes $\mathcal{H}(\mathbf{X})$ continuously into itself follows from Theorem 4.1. To apply Theorem 4.1, three conditions must be met. These conditions are (i), (ii) and (iii), stated next. f is analytic on $\hat{\mathbb{C}}$ so (i) it is continuous, and (ii) it maps open sets to open sets. The way in which $\mathbf{X}$ is constructed ensures that, for small enough ϵ , (iii) $f(\mathbf{X}) \supset \mathbf{X}$. (The latter implies $W(\mathbf{X}) = f^{-1}(\mathbf{X}) \subset \mathbf{X}$.)

To prove that W possesses a unique fixed point we again make use of Theorem 4.1. Consider the limit $A \in \mathcal{H}(\mathbf{X})$ of the decreasing sequence of sets, $\{W^{on}(\mathbf{X})\}$, namely,

$$A = \bigcap_{n=1}^{\infty} f^{o(-n)}(\mathbf{X}) = \lim_{n \rightarrow \infty} W^{on}(\mathbf{X}).$$

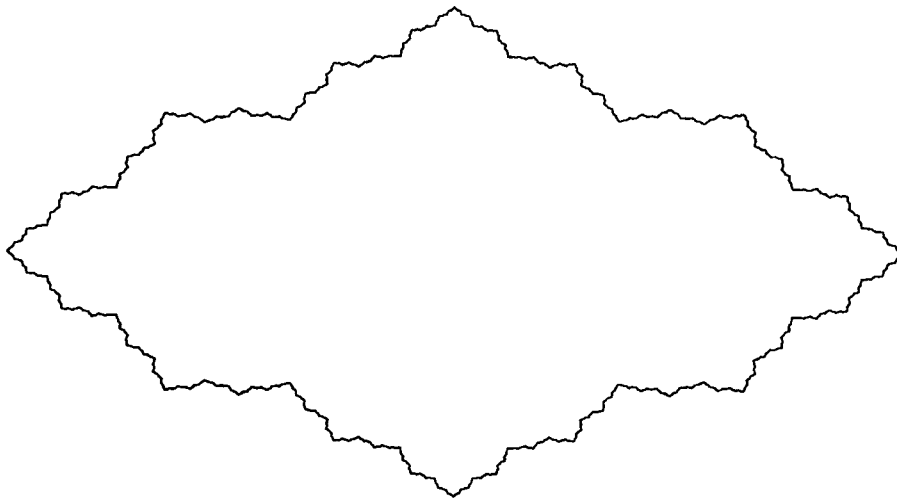


Figure VII.202. (a)–(d) The results of applying the Random Iteration Algorithm to the IFS $\{\hat{\mathbb{C}}; \sqrt{z + \lambda}, -\sqrt{z + \lambda}\}$ for various values of $\lambda \in [0, 3]$. Compare these images with those in Figure VII.198. The results are pictures of the Julia set for $f_\lambda(z) = z^2 - \lambda$. The connection between these Julia sets and IFS theory is revealed!

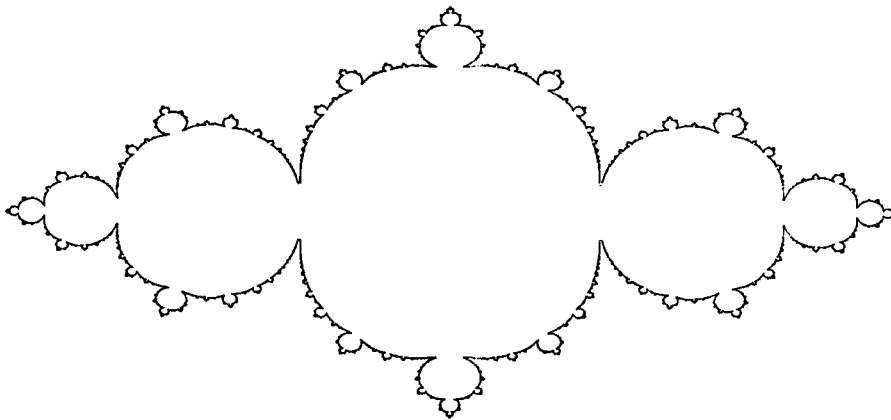


Figure VII.202. (b)

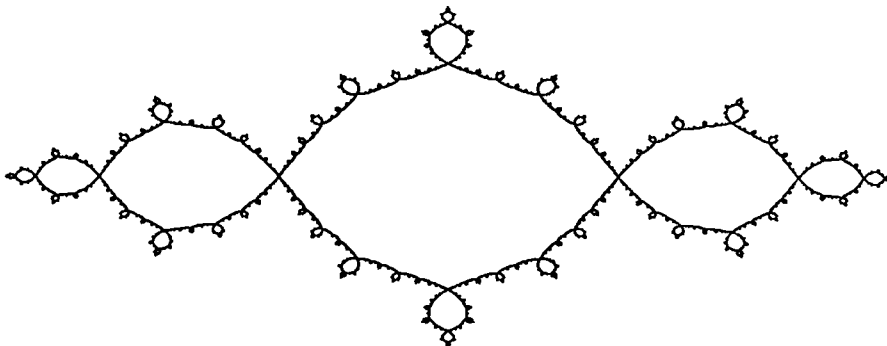


Figure VII.202. (c)

Figure VII.202. (d)

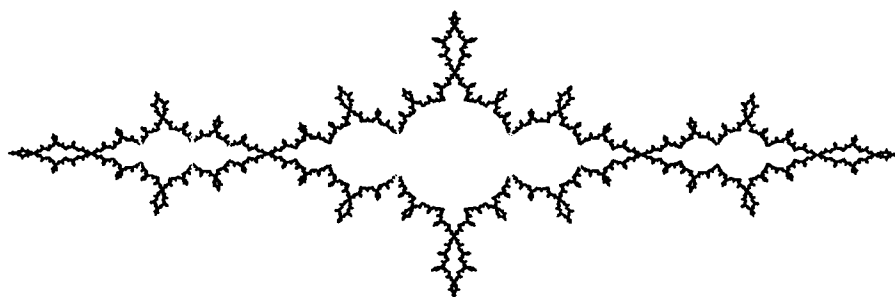
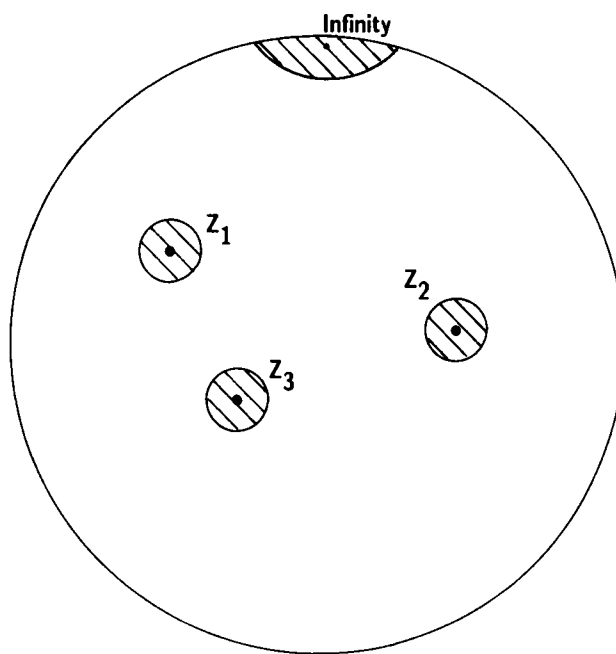


Figure VII.203. The Riemann Sphere $\hat{\mathbb{C}}$ with a number of very small open balls of radius ϵ removed. One ball is centered at each of the points $\{z_p \in \mathbb{C}\}$ belonging to an attractive cycle of the transformation $f_\lambda(z) = z^2 - \lambda$. One ball is centered at the point at infinity.



This obeys $W(A) = A$. It follows from [Brolin 65], Lemma 6.3, that $A = J_\lambda$, the Julia set. This completes the sketch of the proof.

Theorem 2.1 can be generalized to apply to polynomial transformations $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ of degree N greater than 1. Here is a rough description: let $f^{-1}(z) = \{w_1(z), w_2(z), \dots, w_N(z)\}$ denote a definition of branches of the inverse of f . Then consider the IFS $\{\hat{\mathbb{C}}; w_1(z), w_2(z), \dots, w_N(z)\}$. This IFS is not hyperbolic: the “typical” situation is that the associated operator $W : \mathcal{H}(\hat{\mathbb{C}}) \rightarrow \mathcal{H}(\hat{\mathbb{C}})$ possesses a finite number of fixed points, all except one of which are “unstable.” The one “stable” fixed point is J_f , and $W^n(A) \rightarrow J_f$ for “most” $A \in \mathcal{H}(\hat{\mathbb{C}})$. In principle, J_f can be computed using the Random Iteration Algorithm.

Results like Theorem 2.1 are concerned with what are known as *hyperbolic* Julia sets. The Julia set of a rational transformation $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ is hyperbolic if, whenever $c \in \hat{\mathbb{C}}$ is a critical point of f , the orbit of c converges to an attractive cycle of f . The Julia set for $z^2 - 0.75$ is an example of a nonhyperbolic Julia set. We refer to [Peitgen 1986] as a good source of further information about Julia sets from the dynamical systems' point of view.

Explicit formulas for the inverse maps, $\{w_n(z) : n = 1, 2, \dots, N\}$, for a polynomial of degree N , are not generally available. So the Random Iteration Algorithm cannot usually be applied. Pictures of Julia sets and filled Julia sets are often computed with the aid of the Escape Time Algorithm. The case of quadratic transformations is somewhat special, because both algorithms can be used. The Random Iteration Algorithm can also be applied to compute Julia sets of cubic and quartic polynomials, and of special polynomials of higher degree such as $z^n + \lambda$ where $n = 5, 6, 7, \dots$, and $\lambda \in \mathbb{C}$.

Examples & Exercises

2.1. Consider the dynamical system $\{\hat{\mathbb{C}}; f(z) = z^2\}$. The origin, O , is an attractive cycle of period 1: indeed $f(O) = O$ and $|f'(O)| = 0 < 1$. Notice that $\lim_{n \rightarrow \infty} f^{\circ n}(z) = O$ for all $z \in B(O, 0.99999999)$. Let $\mathring{B}(z, r)$ denote the open ball on $\hat{\mathbb{C}}$, with center at z and radius r . Theorem 2.1 tells us that the IFS

$$\{X = \hat{\mathbb{C}} \setminus \{\mathring{B}(O, 0.0000001) \cup \mathring{B}(\infty, 0.0000001)\}; w_1(z) = \sqrt{z}, w_2(z) = -\sqrt{z}\}$$

possesses a unique attractor. The attractor is actually the circle of radius 1 centered at the origin. It can be computed by means of the Random Iteration Algorithm. Notice that if we extend the space X to include O , then $O \in \mathcal{H}(X)$ and $O = W(O) = w_1(O) \cup w_2(O)$. If we extend X to include $\mathring{B}(O, 0.0000001)$, then the filled Julia set F_0 belongs to $\mathcal{H}(X)$ and obeys $F_0 = W(F_0)$. If we take X to be all of $\hat{\mathbb{C}}$ then $\hat{\mathbb{C}} = W(\hat{\mathbb{C}})$. In other words, if the space on which the IFS acts is too large then uniqueness of the "attractor" of the IFS is lost.

Can you find two more nonempty compact subsets of $\hat{\mathbb{C}}$ that are fixed points of W , in the case $X = \hat{\mathbb{C}}$?

Establish that, for all $\lambda \in (-0.25, 0.75)$, the point $z_0 = 0.5 - \sqrt{0.25 + \lambda}$ is an attractive cycle of period 1 for $\{\hat{\mathbb{C}}; z^2 - \lambda\}$. Deduce that the corresponding IFS, acting on a suitably chosen space X , possesses a unique attractor.

2.2. Let $\lambda \in (0.75, 1.25)$. Consider the dynamical system $\{\hat{\mathbb{C}}; f(z) = z^2 - \lambda\}$. Let $z_1, z_2 \in \mathbb{R}$ denote the two solutions of the equation $z^2 + z + (1 - \lambda) = 0$. Show $f(z_1) = z_2, f(z_2) = z_1, |(f^{\circ 2})'(z_1)| = |(f^{\circ 2})'(z_2)| < 1$ and hence that $\{z_1, z_2\}$ is an attractive cycle of period 2. Deduce that the IFS

$$\{\hat{\mathbb{C}} \setminus \{\overset{\circ}{B}(z_1, \epsilon) \cup \overset{\circ}{B}(z_2, \epsilon) \cup \overset{\circ}{B}(\infty, \epsilon)\}; +\sqrt{z+\lambda}, -\sqrt{z+\lambda}\}$$

possesses a unique attractor when ϵ is sufficiently small.

2.3. The Julia set J_λ for the polynomial $z^2 - \lambda$ is a union of two “copies” of itself. Identify these two copies for various values of λ . Explain how, when $\lambda = 1$, the two inverse maps $w_1^{-1}(z)$ and $w_2^{-1}(z)$ rip the Julia set apart, and the set map $W = w_1 \cup w_2$ puts it back together again. Where is the rip? Describe the geometry of what is going on here.

2.4. Consider the one-parameter family of polynomials $f(z) = z^3 - \lambda$, where $\lambda \in \mathbb{C}$ is the parameter. Give explicit formulas for the real and imaginary parts of three inverse functions $w_1(z)$, $w_2(z)$, and $w_3(z)$ such that $f^{-1}(z) = \{w_1(z), w_2(z), w_3(z)\}$ for all $\lambda \in \mathbb{C}$. Compute images of the filled Julia set for $f(z)$ for $\lambda = 0.01$ and $\lambda = 1$. Compare these images with those obtained by applying the Random Iteration Algorithm to the IFS $\{C; w_1(z), w_2(z), w_3(z)\}$.

2.5. Consider the dynamical system $\{\hat{\mathbb{C}}; f(z) = z^2 - \lambda\}$ for $\lambda > 2$. Show that $\{f^{on}(O)\}$ converges to the point at infinity. Deduce that the IFS

$$\{X = \hat{\mathbb{C}} \setminus B(\infty, \epsilon); +\sqrt{z+\lambda}, -\sqrt{z+\lambda}\}$$

possesses a unique attractor $A(\lambda)$. $A(\lambda)$ is a generalized Cantor set. Compute some pictures of $A(3)$. Use the Collage Theorem to help find a pair of affine transformations $w_i : \mathbb{R} \rightarrow \mathbb{R}$, $i = 1, 2$, such that the attractor of the IFS $\{\mathbb{R}; w_1, w_2\}$ is an approximation to $A(3)$. Define $\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}$ by $\tilde{f}(x) = w_1^{-1}(x)$ when $x < 0$ and $\tilde{f}(x) = w_2^{-1}(x)$ when $x \geq 0$. Compare the graphs of the functions $f(x) = x^2 - 3$ and $\tilde{f}(x)$ for, say, $x \in [-4, 4]$. Compare one-dimensional “images” obtained by applying the Escape Time Algorithm in a similar manner to both $\{\mathbb{R}; f\}$ and $\{\mathbb{R}; \tilde{f}\}$.

One can sometimes obtain a *hyperbolic* IFS associated with a Julia set, if the domains and ranges of the inverse transformations are defined carefully. The following theorem provides such an example.

Theorem 2.2 *Let $\lambda \in [-0.249, 0.749]$, and let ϵ be a very small positive number. Let $a = 0.5 - \sqrt{0.25 + \lambda}$, an attractive fixed point of the dynamical system $\{\hat{\mathbb{C}}; f(z) = z^2 - \lambda\}$. Let $\tilde{X} = \hat{\mathbb{C}} \setminus \{\overset{\circ}{B}(a, \epsilon) \cup \overset{\circ}{B}(\infty, \epsilon) \cup (0, \infty)\}$. That is, $\tilde{X}$ consists of the Riemann Sphere with a small open ball centered at a , a small open ball centered at ∞ , and the open interval $(0, \infty)$, removed. (This space is not compact because the edges of the lips of the cut, from O to ∞ , are missing.) To each lip let there be attached copies of the pieces of the real interval $(0, \infty)$ which were removed, to provide a compact space X , as illustrated in Figure VII.204. The distance $d(z_1, z_2)$ between a pair of points z_1 and $z_2 \in X$ is the length (measured using the spherical metric) of the shortest path that lies in X and connects z_1 to z_2 . (Paths in X cannot cross the cut, they have to go around it.) (X, d) is a compact metric space.*

Define $w_1 : X \rightarrow X$ by $w_1(z) = \sqrt{z+\lambda}$, the root that lies in the “upper half

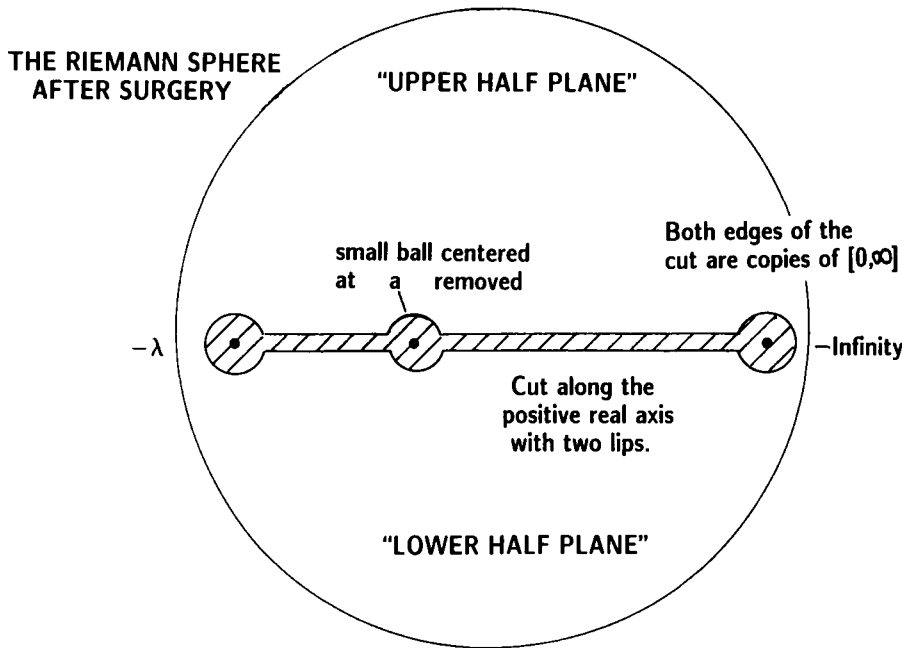


Figure VII.204. Construction of a compact metric space X for IFS $\{X; \sqrt{z + \lambda}, -\sqrt{z + \lambda}\}$ with $\lambda \in (-0.25, 0.75)$, used in Theorem 2.2. Both sides of the “slit” from $-\lambda$ to infinity belong to the space. The distance between a pair of points on X is the length of the shortest path that connects the points without crossing the slit. The distance between points may be much greater than it looks.

plane.” For z on the upper edge of the cut, $w_1(z)$ is also on the upper edge. For z on the lower edge of the cut, $w_1(z)$ lies on the negative real axis. Define $w_2 : X \rightarrow X$ by $w_2(z) = -\sqrt{z + \lambda}$, the root that lies in the “lower half plane.” For z on the upper edge of the cut, $w_2(z)$ lies on the negative real axis; and, for z located on the lower edge of the cut, $w_2(z)$ lies on the lower edge of the cut.

Then there is a metric on X , equivalent to the metric d , such that the IFS $\{X; w_1, w_2\}$ is hyperbolic. The attractor is the Julia set J_λ for $z^2 - \lambda$, where the real point $0.5 + \sqrt{0.25 + \lambda}$ is repeated on both the upper and the lower edges of the cut.

Sketch of proof Let $e = (e_1, e_2, \dots, e_n, \dots) \in \Sigma$, the code space on the two symbols $\{1, 2\}$. Define a sequence of nonempty compact subsets of X by

$$X_n(e) = w_{e_1} \circ w_{e_2} \circ w_{e_3} \circ w_{e_4} \circ w_{e_5} \circ w_{e_6} \circ \dots \circ w_{e_n}(X) \text{ for } n = 1, 2, 3, \dots$$

It follows, using [Brolin], Theorem 6.2, and Lemma 6.3, that the sequence $\{X_n \in \mathcal{H}(X)\}$ converges to a singleton, say $\{\phi(e)\}$, where $\phi(e) \in J_\lambda$ and

$$\bigcup_{e \in \Sigma} \phi(e) = J_\lambda.$$

A beautiful theorem of Elton [Elton 1988] applies under just these conditions and provides the conclusion of the theorem. This completes the outline of the proof.

In those situations where the IFS $\{X; +\sqrt{z + \lambda}, -\sqrt{z + \lambda}\}$ is hyperbolic one can use the associated code space to discuss both the Julia set and the associated shift

dynamical system $\{J_\lambda; f(z) = z^2 - \lambda\}$. Here we give some of the flavor of such a discussion. More details can be found in [Barnsley 1984].

For the remainder of this section let $\lambda \in (-0.25, 0.75)$ and consider the IFS $\{\mathbf{X}; w_1(z) = \sqrt{z + \lambda}, w_2(z) = -\sqrt{z + \lambda}\}$, as defined in Theorem 2.2. Let Σ denote the code space on the two symbols $\{1, 2\}$, and let $\phi: \Sigma \rightarrow J_\lambda$ denote the associated code space map, introduced in Theorem 2.1 in Chapter IV. If $e = (e_1, e_2, \dots, e_n, \dots) \in \Sigma$, then

$$\phi(e) = \lim_{n \rightarrow \infty} w_{e_1} \circ w_{e_2} \circ w_{e_3} \circ w_{e_4} \circ w_{e_5} \circ w_{e_6} \circ \dots \circ w_{e_n}(z).$$

Replace the symbol “1” by the symbol “+” and replace the symbol “2” by the symbol “−”. Then the point $\phi(e)$ on the Julia set J_λ can be represented by the formula

$$\phi(e) = e_1 \sqrt{\lambda e_2 \sqrt{\lambda e_3 \sqrt{\lambda e_4 \sqrt{\lambda e_5 \sqrt{\lambda e_6 \sqrt{\lambda e_7 \sqrt{\lambda e_8 \dots e_n \sqrt{\lambda \dots}}}}}}}}},$$

where $e_i \in \{+, -\}$ for each positive integer i . The set J_λ itself can be represented by the collection of formulas

$$\pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \dots \pm \sqrt{\lambda \dots}}}}}}}}}, \quad (1)$$

where all possible sequences of plus and minus signs are permitted. A particular sequence of signs, corresponding to a point in J_λ , is an address of the point.

In Figure VII.205 we show the Julia set for $z^2 - 0.7$, with the addresses of various points marked on it. Some points on $J_{0.7}$ have multiple addresses while others have single addresses. It appears that the the IFS is just-touching.

The shift dynamical system associated with the IFS is $\{J_\lambda; f(z) = z^2 - \lambda\}$. Notice how the set of points represented by the formulas in equation 1 is mapped into itself by the function that “squares” a formula and subtracts λ from the result. A point on a cycle of period 2 is represented by

$$+ \sqrt{\lambda - \sqrt{\lambda + \sqrt{\lambda - \sqrt{\lambda + \sqrt{\lambda - \sqrt{\lambda + \sqrt{\lambda - \dots + \sqrt{\lambda \dots}}}}}}}}.$$

The other point on this cycle is obtained by squaring the formula and subtracting λ .

In Theorem 2.4 in Chapter IV, we learned that the set of periodic points of the shift dynamical system associated with a hyperbolic IFS is dense in the attractor of the IFS. Here this tells us that the set of periodic points of the dynamical system

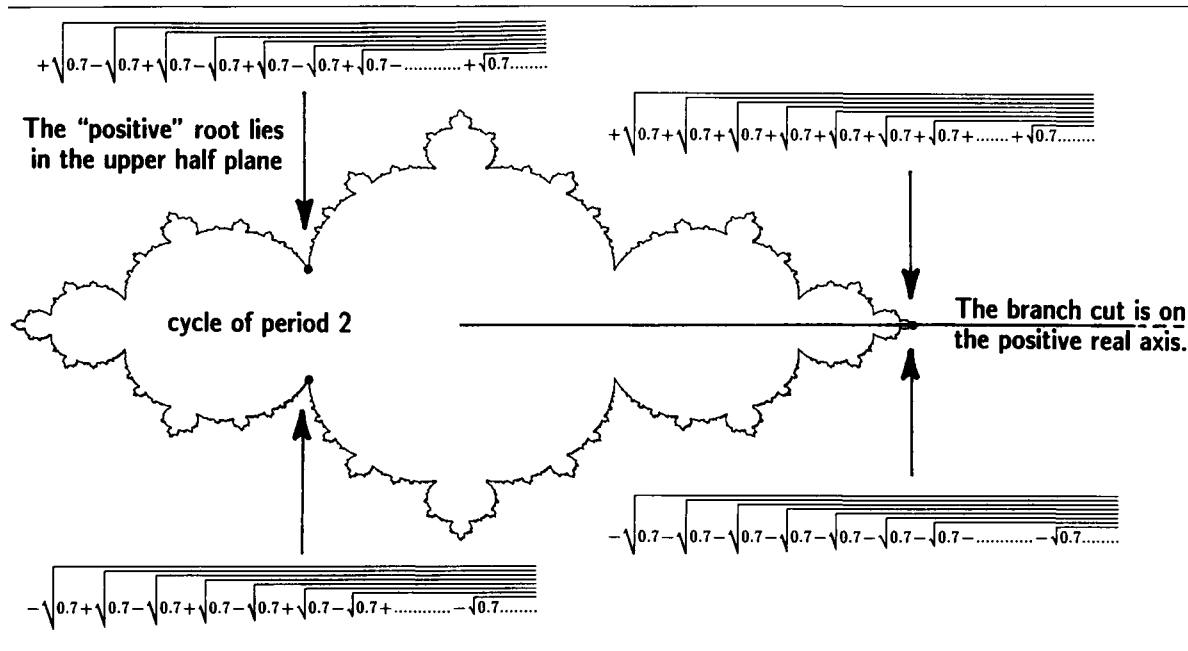


Figure VII.205. The Julia set for $z^2 - 0.7$, labelled with various addresses. Chaotic dynamics takes place on the Julia set and orderly dynamics takes place off it. Boundaries of a fractal character often separate regions where the dynamical system behaves differently. The behavior of the dynamical system on such a boundary may then be indecisive, and in some way chaotic.

$\{J_\lambda; f(z) = z^2 - \lambda\}$ is dense in J_λ . In fact, a related idea was the starting point of Julia's original investigations. He considered dynamical systems of the form $\{\hat{\mathbb{C}}; f(z)\}$, where $f(z)$ is analytic. He defined the (Julia) set to be the closure of the set of repulsive cycles of f .

Following Theorem 8.1 in Chapter IV we explained the sense in which the shift dynamical system associated with a hyperbolic IFS is chaotic. In the present context we learn that the dynamical system $\{J_\lambda; z^2 - \lambda\}$ is chaotic.

One can think of the dynamical system $\{\hat{\mathbb{C}}; z^2 - \lambda\}$ as being the union of two dynamical systems, a chaotic one $\{J_\lambda; z^2 - \lambda\}$ and an orderly one $\{\hat{\mathbb{C}} \setminus J_\lambda; z^2 - \lambda\}$. The orbit of any point in the latter system converges to a fixed point of the transformation. The orbits of "most" points in the former system are wild. In practice they are usually so wild they cannot be constrained to remain on the repelling set J_λ . They escape and thereafter behave in a rather predictable manner.

An example of chaotic dynamics on a Julia set is provided by the dynamical system $\{[0, 1]; f(x) = 4x(1 - x)\}$. The interval $[0, 1]$ is exactly the Julia set for the transformation. This system is close to the "chaotic" one illustrated in Figure IV.108.

Examples & Exercises

2.6. The Julia set for $z^2 - 2$ is the interval $[-2, 2]$. Show that the shift dynamical system associated with the IFS $\{[-2, 2]; +\sqrt{z+2}, -\sqrt{z+2}\}$ is precisely the dynamical system $\{[-2, 2]; z^2 - 2\}$. Use a chain of square roots to locate a cycle of minimal period 3.

2.7. Verify numerically that for various choices of $\pm$ on each square root, and for various complex numbers λ such that $|\lambda|$ is very small, the expression below evaluates approximately to a complex number that lies on the unit circle centered at the origin, if enough square roots are taken. ($+\sqrt{z}$ means the solution, w , of the equation $w^2 = z$, which lies either on the nonnegative real axis or in the upper half plane.) Make a hand-waving explanation of why this is, in terms of Julia set theory.

$$\pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \sqrt{\lambda \pm \dots \pm \sqrt{\lambda}}}}}}}}$$

2.8. Design an IFS with condensation such that its attractor looks like an infinite nested chain of square root signs.



2.9. Figure VII.206 represents a sequence of sets $\{A_n\}$ which converges to the Julia set of $f(z) = z^2 - 1$. A_0 denotes the union of the two largest faces and $A_n = f^{\circ(-n)}(A_0)$. Identify the set A_2 .

3 The Application of Julia Set Theory to Newton's Method

We are familiar, since our first course in calculus, with Newton's method for computing solutions of the equation $F(x) = 0$. Or are we?

Consider the polynomial $F(z) = z^4 - 1$ for $z \in \mathbb{C}$. There are four distinct complex numbers, a_i ($i = 1, 2, 3, 4$) such that $F(a_i) = 0$. These are called the roots, or the zeros, of the polynomial $F(z)$. Newton's method provides a means to compute them. Pretend that we do not know that $a_1 = 1$, $a_2 = -1$, $a_3 = i$, and $a_4 = -i$. Then Newton tells us to consider the dynamical system

$$\{\hat{\mathbb{C}}; f(z) = z - \frac{F(z)}{F'(z)}\}.$$

We call $f(z)$ the *Newton transformation* associated with the function $F(z)$. The general expectation is that a typical orbit $\{f^{\circ n}(z_0)\}$, which starts from an initial

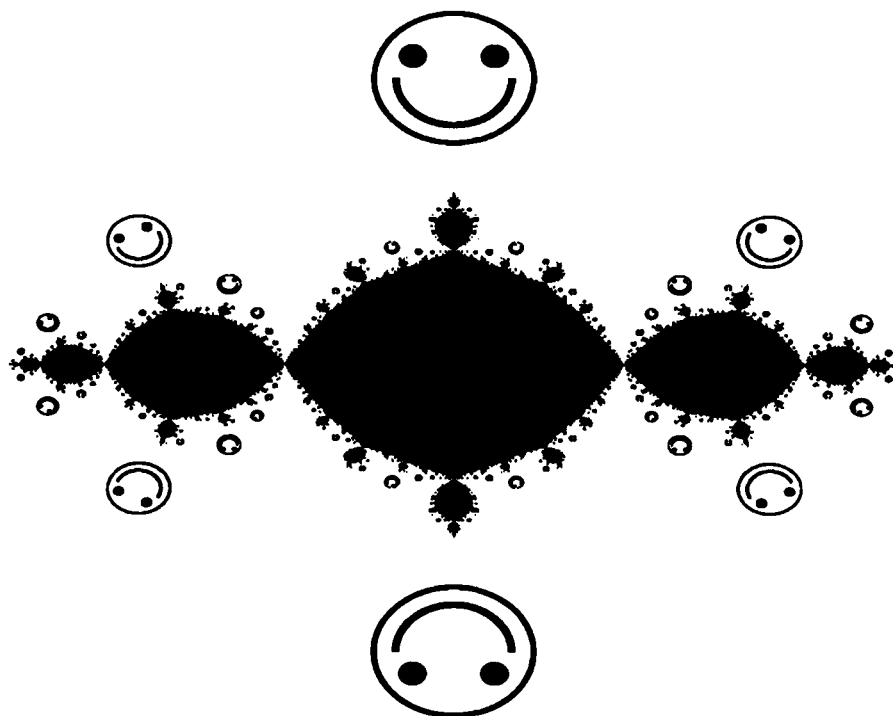


Figure VII.206. Let $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ denote the polynomial $z^2 - 1$. Let A_0 be the union of the two largest smiley faces. Define a transformation $W : \mathcal{H}(\hat{\mathbb{C}}) \rightarrow \mathcal{H}(\hat{\mathbb{C}})$ by $W(B) = f^{-1}(B)$ for all $B \in \mathcal{H}(\hat{\mathbb{C}})$ and let $A_n = W^{on}(A_0)$ for $n = 0, 1, 2, \dots$. Then the sequence of sets $\{A_n\}$ converges to the Julia set $z^2 - 1$. Can you identify A_2 ?

“guess” $z_0 \in \mathbb{C}$, will converge to one of the roots of $F(z)$. In the present example, the Newton transformation is given by

$$f(z) = \frac{3z^4 + 1}{4z^3}.$$

We expect the orbit of z_0 to converge to one of the numbers a_1, a_2, a_3 , or a_4 . If we choose z_0 close enough to a_i , then it is readily proved that

$$\lim_{n \rightarrow \infty} f^{on}(z_0) = a_i, \quad \text{for } i = 1, 2, 3, 4.$$

If, on the other hand, z_0 is far away from all of the a_i 's, then what happens? Perhaps the orbit of z_0 converges to the root of $F(z)$ closest to z_0 ? Or perhaps the orbit does not settle down, but wanders, hopelessly, forever?

Let us make a computergraphical experiment to help answer these questions. We use the Escape Time Algorithm to produce a picture of those points $z_0 \in \hat{\mathbb{C}}$ whose orbits converge to a_1 . Define $\mathcal{W} = \{(x, y) \in \mathbb{C} : -2 \leq x \leq 2, -2 \leq y \leq 2\}$ and $\mathcal{V} = \{z \in \mathbb{C} : |z - a_1| \leq 0.0001\}$. The real and imaginary parts of $f(x + iy)$ are given by

$$f(x + iy) = \frac{(ce + df)}{(e^2 + f^2)} + i \frac{(de - cf)}{(e^2 + f^2)},$$

where $a = x^2 - y^2$, $b = 2xy$, $c = 3a^2 - 3b^2 + 1$, $d = 6ab$, $e = 4(xa - yb)$, and $f = 4(xb + ya)$. Program 7.1.1 is modified accordingly. Pixels corresponding to points in $\mathcal{W}$ whose orbits reach $\mathcal{V}$ in less than a fixed number of iterations are plotted. A picture resulting from such an experiment is shown in Figure VII.207. See also Figures VII.208 and VII.209.

Color Plate 14 shows the output from another such experiment. This time Mercator's projection is used to represent the Riemann sphere, and points whose orbits converge to the different points a_1 , a_2 , a_3 , and a_4 are plotted in different colors.

The following definition is equivalent to Definition 1.1 in the case of polynomials, [Brolin].

Definition 3.1 *The Julia set of a rational function $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$, of degree greater than 1, is the closure of the set of repulsive periodic points of the dynamical system $\{\hat{\mathbb{C}}; f\}$.*

For the rational function $f(z)$ considered above, one can prove that the Julia set J is the same as the set of points whose orbits do not converge to any one of the points a_1 , a_2 , a_3 , a_4 . In Figure VII.207, $J \cap \mathcal{W}$ is represented by the boundary between the black and white regions. In Color Plate 14, $J \cap \mathcal{W}$ is the place where the four colors meet. The complement of the Julia set consists of four open sets, the *basins of attraction* of the four attractive fixed points of the Newton iteration scheme. In Color Plate 14 the dark green region represents part of the basin of attraction of a_1 . The black regions in the color plate are caused by (a) rounding errors, and (b) the fact that only 100 points on each orbit are tested, for convergence to one of the points a_1 , a_2 , a_3 , a_4 .

The Julia set J is the part of $\hat{\mathbb{C}}$ on which chaotic dynamics occurs. It can be characterized as the closure of the set of points whose orbits wander, hopelessly, forever. Orderly, slightly boring motion takes place on $\hat{\mathbb{C}} \setminus J$. J is the boundary of the white region. It is the boundary of the dark green region. It is a bonafide fractal, yet nobody knows its fractal dimension.

There is a beautiful theorem of Sullivan, which can be illustrated using the “petals” in Color Plate 14. The complement of the Julia set is the union of a countable collection of connected open sets, which we call petals. If P is a petal then $f(P)$ is another petal. The Non-Wandering Domain Theorem [Sullivan 1982] says that no connected component of the complement of the Julia set wanders, hopelessly, forever. It always settles into a periodic orbit of petals. If P is a petal in the present example, then one can prove that there is a positive integer S so that $f^{\circ S}(P) = f^{\circ(S+1)}(P) = f^{\circ(S+2)}(P) = f^{\circ(S+3)}(P) = \dots$. The final petal $f^{\circ S}(P)$ is one of the connected components of the complement of the Julia set that contains one of the points a_1 , a_2 , a_3 , a_4 . Each petal is eventually periodic. The orbit of a petal ends up in a cycle of petals of period 1.

How are we to think about this fabulous Julia set? IFS theory provides a simple point of view, as we show next. We begin by defining the inverse map associated

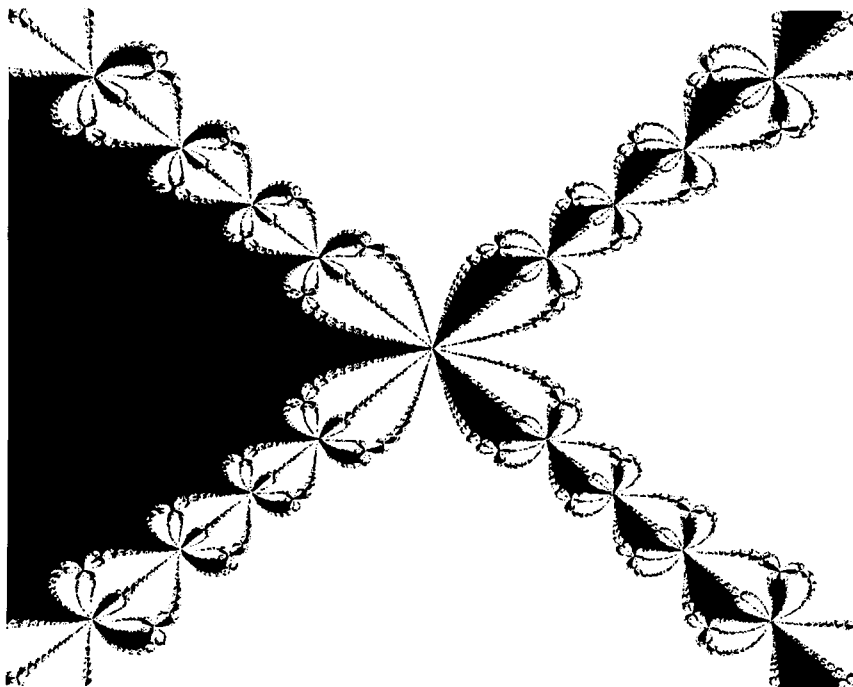


Figure VII.207. The Escape Time Algorithm is applied to analyze Newton's method for finding the complex roots of the polynomial $z^4 - 1$. The boundary of this region represents the Julia set for the rational function $f(z) = (3z^4 + 1)/4z^3$. The points plotted black are those points $z = x + iy$ with $-2 \leq x \leq 2$, and $-2 \leq y \leq 2$, whose orbits intersect $\mathcal{V} = \{z \in \mathbb{C} : |z + 1| \leq 0.0001\}$ in less than 1,000 iterations.

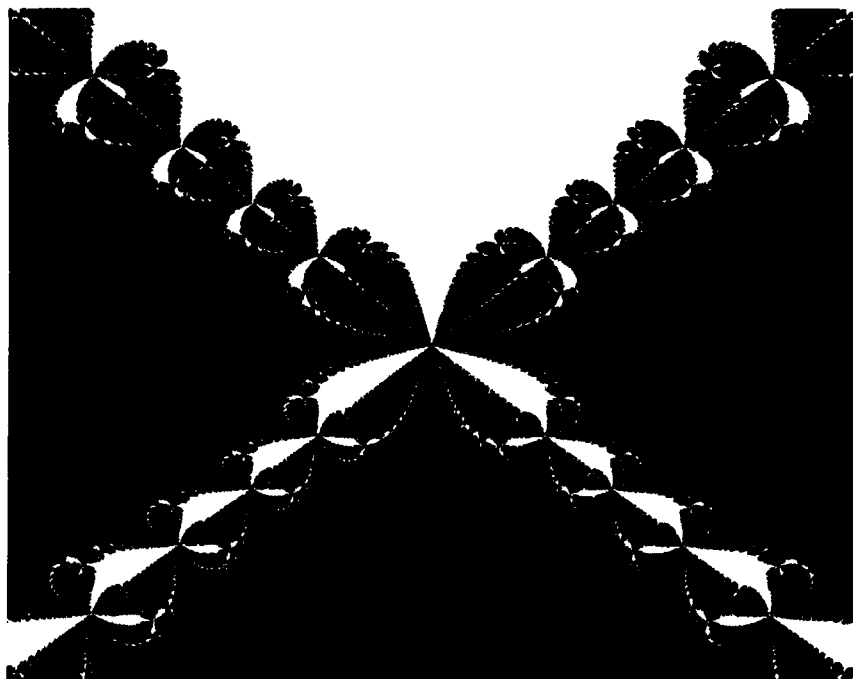
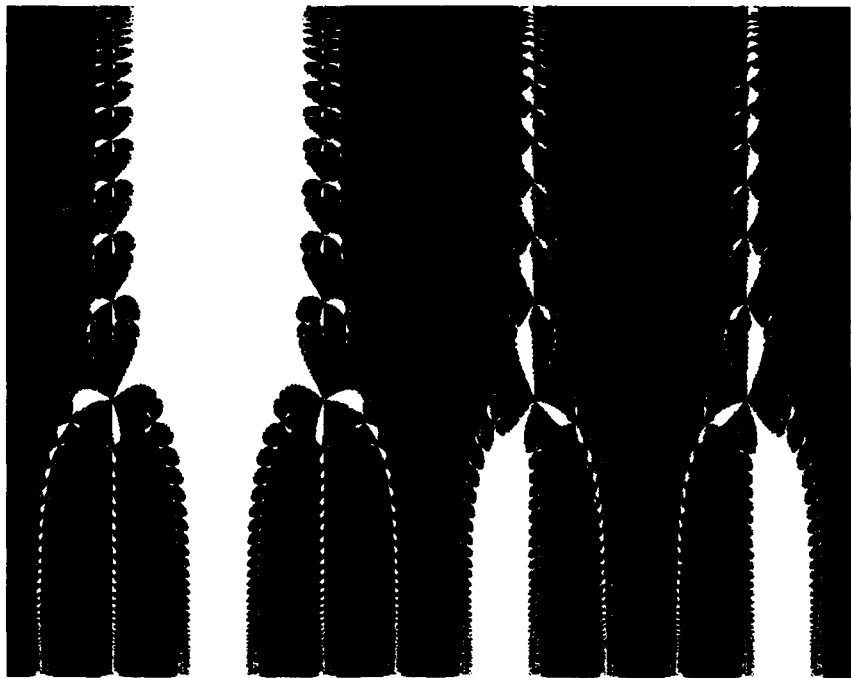


Figure VII.208. The boundary of this region represents the Julia set for the rational function $f(z) = (3z^4 + 1)/4z^3$. The two shades of gray, black and white, correspond to the basins of attraction of the four attractive fixed points of $f(z)$. To which point in $\mathbb{C}$ do the orbits of points in the white region converge?

Figure VII.209. Mercator's projection of the Riemann sphere showing the basins of attraction of the four attractive fixed points of the Newton transformation of $z^4 - 1$. The top of the rectangle corresponds to the point at infinity, and the bottom of the box corresponds to the origin. The points -1 , $+i$, $+1$, and $-i$ lie on the equator. The shading follows the same convention as in (a) and (b). Which point on $\hat{\mathbb{C}}$ is represented by the midpoint of this image?



with f . Let $z \in \hat{\mathbb{C}}$ be given and solve

$$z = \frac{3w^4 + 1}{4w^3},$$

to find w in terms of z . This leads to the quartic equation

$$3w^4 - 4zw^3 + 1 = 0.$$

This has four solutions, when we count solutions according to their multiplicities. We can organize these solutions to provide four functions; that is, we write $f^{-1}(z) = \{w_1(z), w_2(z), w_3(z), w_4(z)\}$. Then the Julia set is the “attractor” for the IFS $\{\hat{\mathbb{C}}; w_i, i = 1, 2, 3, 4\}$. However, as in the case of quadratic transformations on $\hat{\mathbb{C}}$, this statement must be treated cautiously: for example, clearly this IFS admits more than one invariant set.

Theorem 3.1 *Let $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ be the Newton transformation associated with the polynomial $z^4 - 1$. Let $\epsilon > 0$ be very small. Let $X = \hat{\mathbb{C}} \setminus \bigcup_{i=1}^4 \mathring{B}(a_i, \epsilon)$ where $a_1 = 1$, $a_2 = -1$, $a_3 = i$, and $a_4 = -i$. As above, define $W : \mathcal{H}(X) \rightarrow \mathcal{H}(X)$ by*

$$W(B) = \bigcup_{i=1}^4 w_i(B) = f^{-1}(B) \quad \text{for all } B \in \mathcal{H}(X).$$

Then W is continuous, possesses a unique fixed point, J , the Julia set of f , and

$$\lim_{n \rightarrow \infty} W^{on}(B) = J \quad \text{for all } B \in \mathcal{H}(X).$$

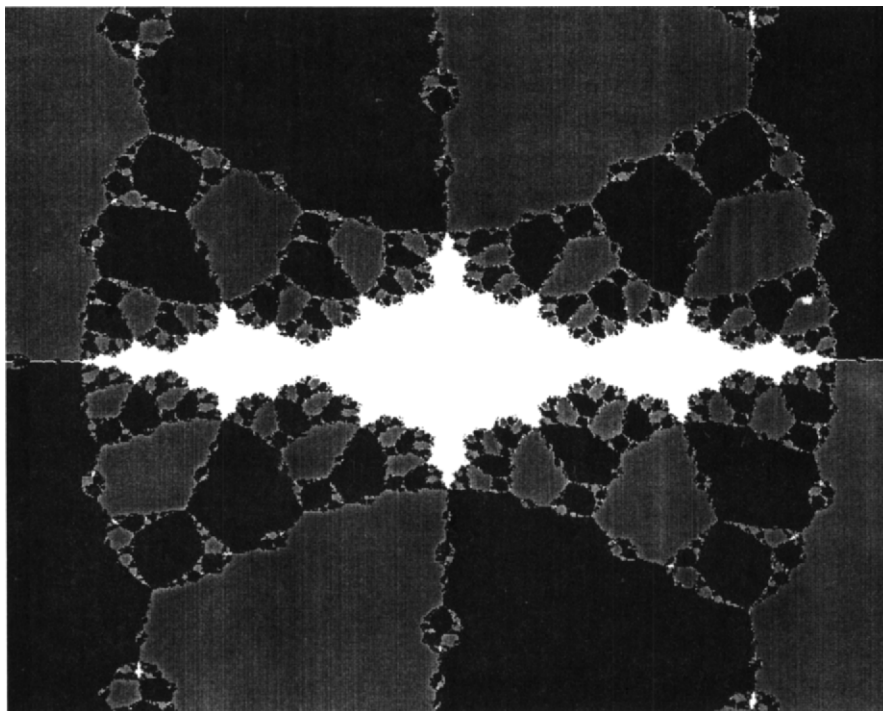


Figure VII.210. The Escape Time Algorithm is applied to a Newton transformation $f(z)$ associated with a cubic polynomial. $f(z)$ possesses an attractive cycle $\{b_1, b_2\}$ of minimal period 2. The basin of attraction of the two-cycle is represented in white. Points whose orbits arrive within a distance of 0.01 of the cycle, prior to 100 iterations, are plotted in white. Does the basin of attraction of the cycle look familiar?

Sketch of proof This is essentially the same as the sketch of the proof of Theorem 2.1.

The Newton transformation associated with a polynomial may possess an attractive cycle of minimal period greater than 1. This cycle may not be directly related to the roots of the polynomial. As an example consider the Newton transformation $f(z)$ associated with the polynomial

$$F(z) = z^3 + (\lambda - 1)z + 1.$$

$\lambda \in \mathbb{C}$ can be chosen so that $f(z)$ possesses an attractive cycle $\{b_1, b_2\}$ of minimal period 2. Figure VII.210 illustrates the basin of attraction of the cycle. The Escape Time Algorithm was used to obtain this image. Points whose orbits arrive within a distance 0.01 of the cycle, prior to 100 iterations, are plotted in black. Accordingly, the escape region is $\mathcal{V} = B(b_1, 0.00001) \cup B(b_2, 0.00001)$. Notice the resemblance of the basin of attraction of $\{b_1, b_2\}$ to the filled Julia set for $z^2 - 1$. This similarity is not accidental. It can be explained using the theory of “polynomial-like” mappings [Douady 1985].

Some interesting computergraphical experiments involving Julia sets for Newton's method are described in [Curry 1983], [Peitgen 1986], and [Vrscay 1986].

Examples & Exercises

3.1. Verify that $z = 1$ is an attractive fixed point for the Newton transformation associated with $F(z) = z^4 - 1$.

3.2. The Newton transformation associated with the polynomial $F(z) = z^2 + 1$ is

$$f(z) = \frac{1}{2} \left(z - \frac{1}{z} \right).$$

Show the corresponding IFS is $\{\hat{\mathbb{C}}; w_1(z) = z + \sqrt{z^2 + 1}, w_2(z) = z - \sqrt{z^2 + 1}\}$, where the square root is defined appropriately. Verify that $A = \mathbb{R} \cup \{\infty\}$ is an attractor of the IFS. Prove, or give evidence to show, that A is the Julia set for $f(z)$. (Hint: see exercise 3.7.) How could the space $\hat{\mathbb{C}}$ be modified so that the IFS has a unique attractor? Notice that numerically computed orbits of points on A , under the dynamical system $\{\hat{\mathbb{C}}; f\}$, can be constrained from escaping from A by keeping imaginary parts equal to zero. Verify numerically that the dynamics of $\{A; f\}$ are wild.

3.3. Find the Newton transformation $f(z)$ associated with the polynomial $F(z) = z^3 - 1$. Use the Escape Time Algorithm to obtain an image, analogous to Figure VII.207, which illustrates this Julia set. Discuss the dynamics of the “petals” in the image.

3.4. In this example we speculate on the application of fractal geometry to biological modelling. Let $F_\lambda(z) = (z - i\lambda)(z - 1)(z + 1)$, where λ is a real parameter, and let $f_\lambda(z)$ denote the associated Newton transformation. Let J_λ denote the Julia set for $f_\lambda(z)$. In Figure VII.211 we show images relating to J_λ , for an increasing sequence of values of λ . These images were computed by applying the Escape Time Algorithm to f_λ .

These images show complex blobs that are reminiscent of something small, biological, and organic. They make one think of the nuclei of cells; of collections of cells during the early stages of development of an embryo; of the process of cell division; and of protozoans. As we track the images we see that the blobs pass through one another. Somehow they do so while preserving their complex geometries. Their geometries seem to interact with one another. Such images suggest that fractal geometry can do more than provide a means for modelling static biological structures, such as ferns: it appears feasible to construct deterministic fractal models, which describe the processes of physiological change that occur during the growth, metamorphosis, and movement of living organisms.

3.5. Find the Newton transformation $f(z)$ associated with the function $F(z) = e^z - 1$. What are the attractive fixed points of the dynamical system $\{\mathbb{C}; f\}$? Figure VII.212 was computed using the Escape Time Algorithm applied to $f(z)$, with $\mathcal{W} = \{(x, y) \in \mathbb{R}^2 : -2.5 \leq x \leq 2.5, -2.5 \leq y \leq 2.5\}$. Describe the main features of the image. Explain, roughly, the causes of some of these features.

3.6. What are the “petals” in the case of the Julia set for $z^2 - 1$? Use a picture of

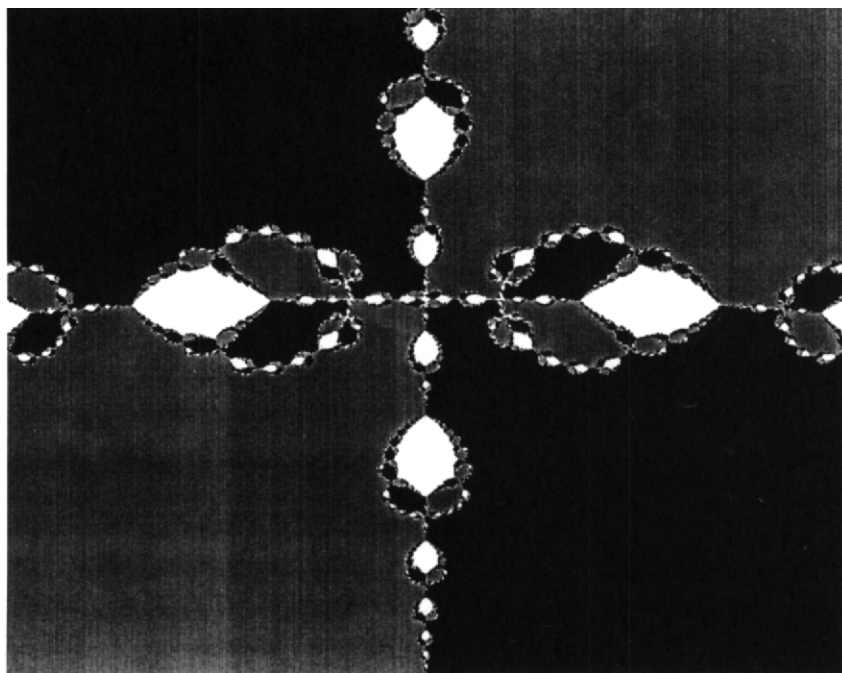


Figure VII.211. (a)–(h) Julia sets associated with a one-parameter family of dynamical systems. Can such systems be used to model biological processes such as meiosis?

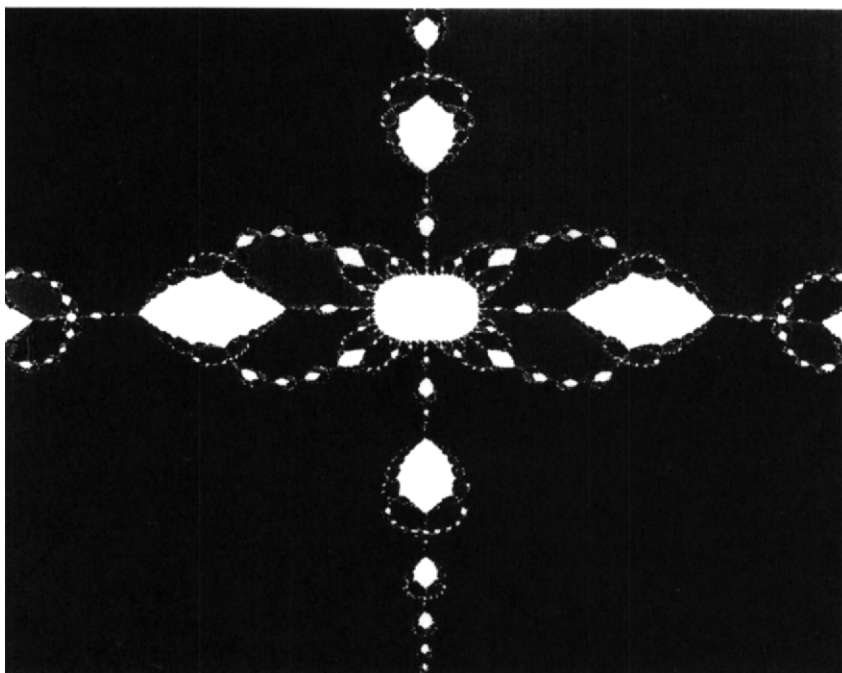


Figure VII.211. (b)

Figure VII.211. (c)

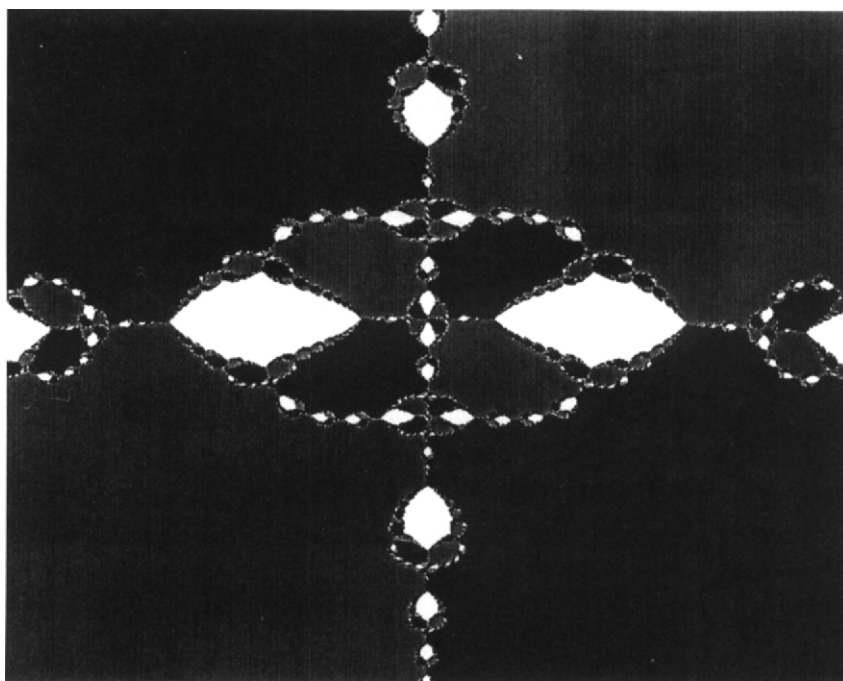
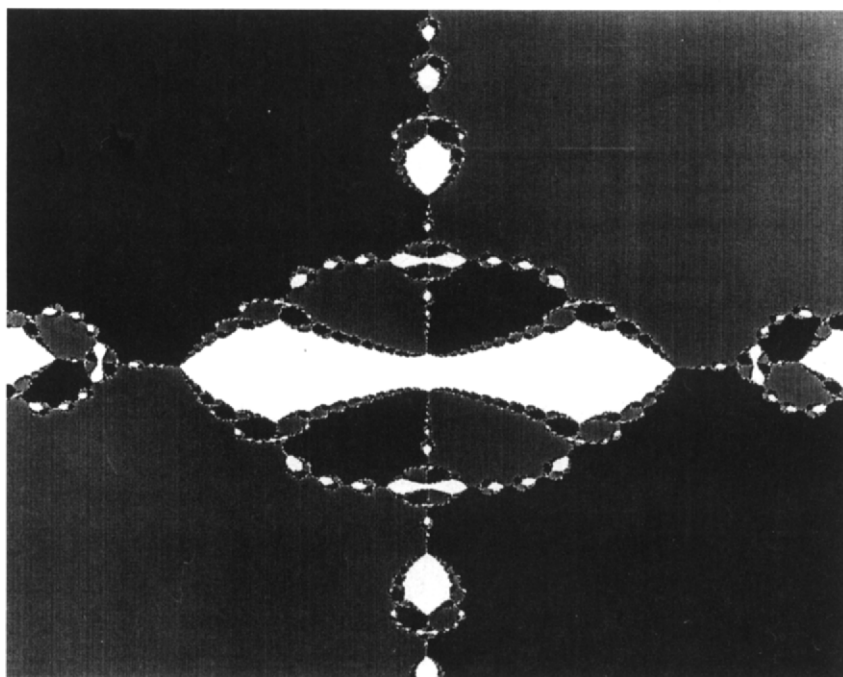


Figure VII.211. (d)



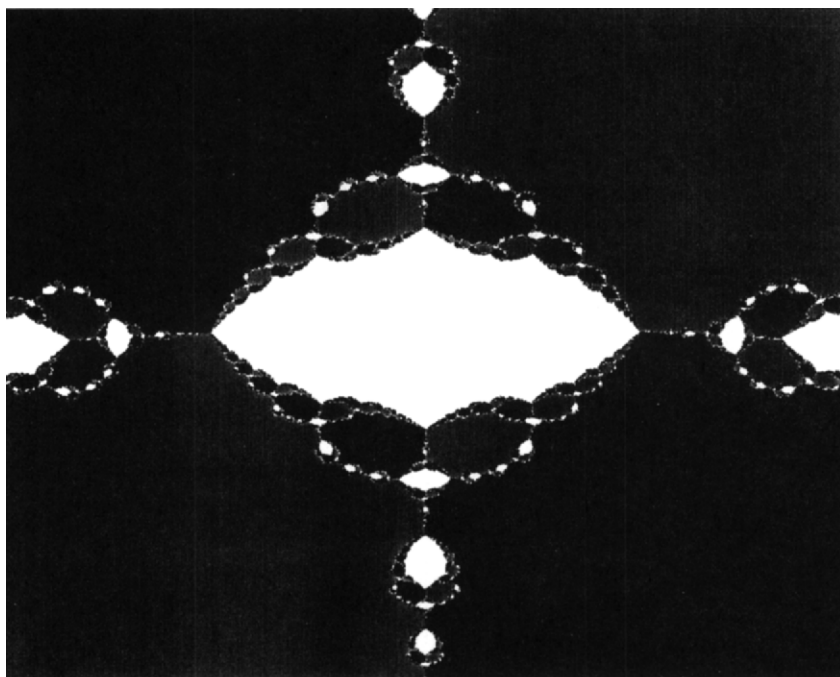


Figure VII.211. (e)

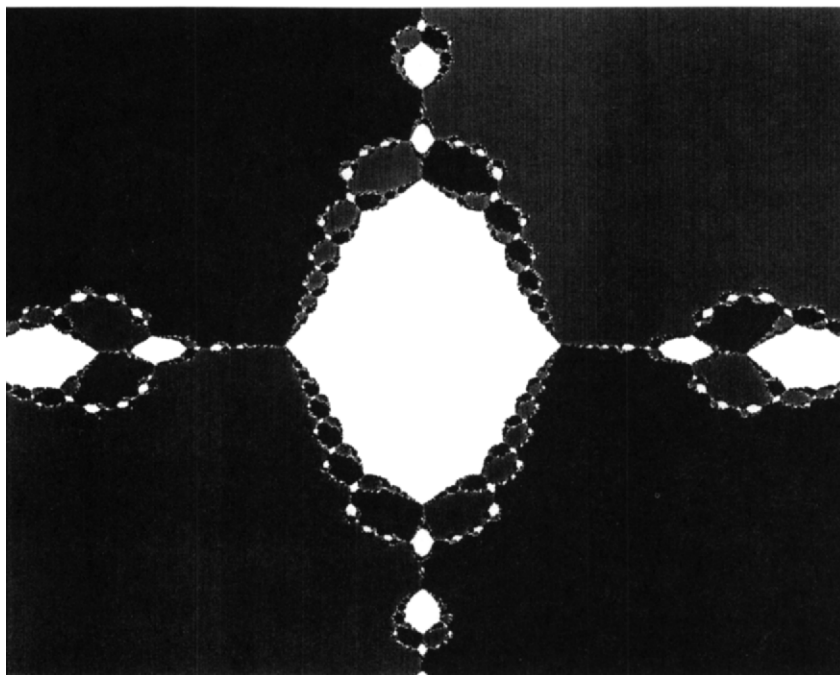


Figure VII.211. (f)

Figure VII.211. (g)

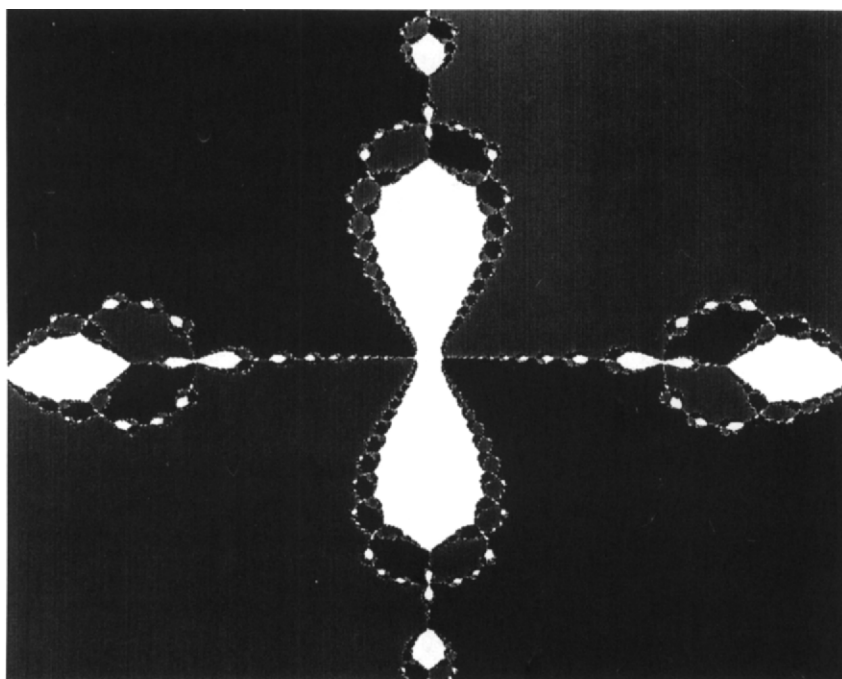
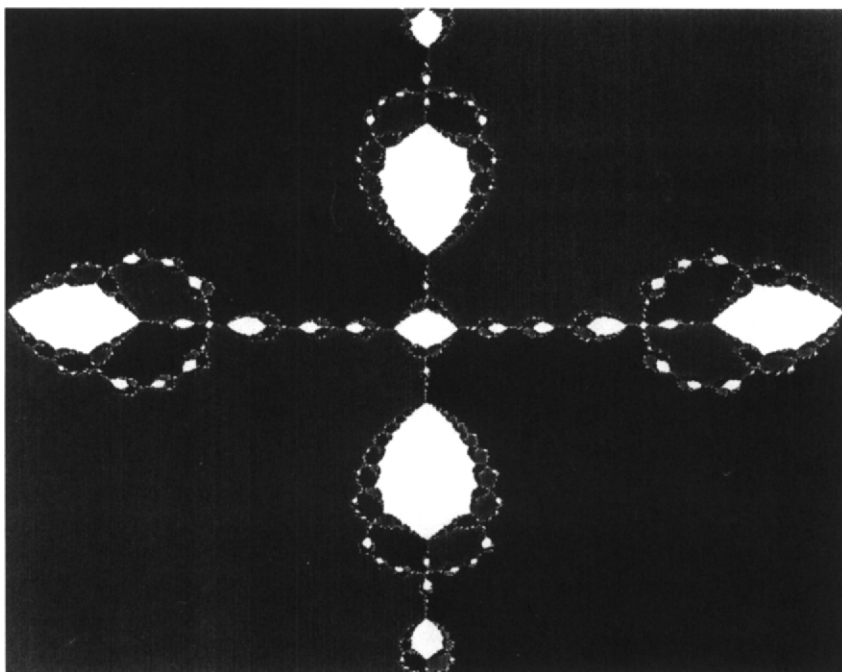


Figure VII.211. (h)



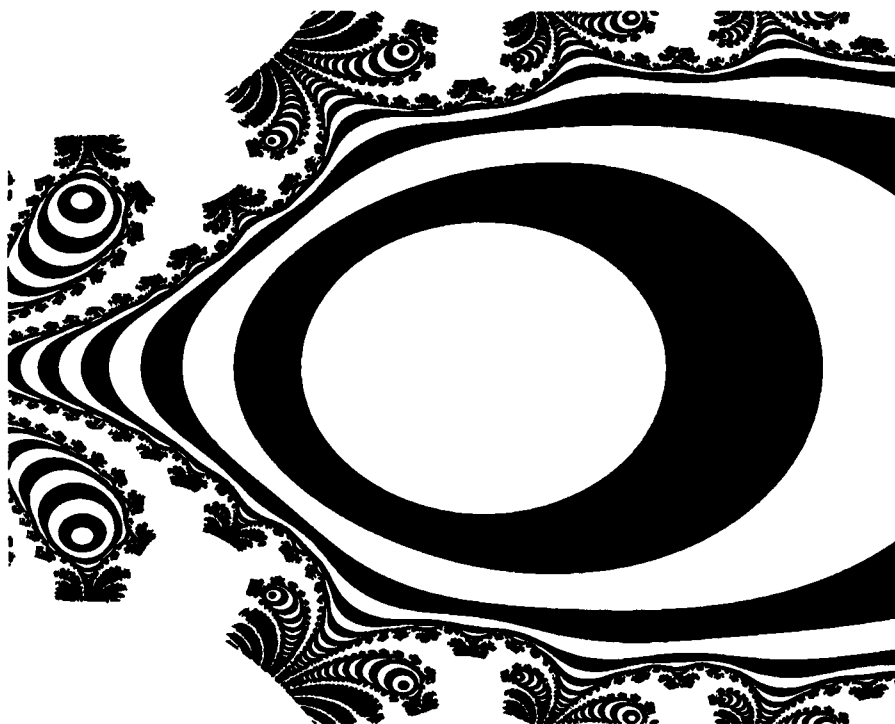


Figure VII.212. This image was computed using the Escape Time Algorithm applied to the Newton transformation associated with $f(z) = e^z - 1$. The viewing window is $\mathcal{W} = \{(x, y) \in \mathbb{R}^2 : -2.5 \leq x \leq 2.5, -2.5 \leq y \leq 2.5\}$. Can you work out what “escape region” was used?

the Julia set for $z^2 - 1$ to illustrate the orbit of a tiny petal that is eventually periodic with minimal period 2.

3.7. By making an explicit change of the coordinates, using a Möbius transformation, show that the following two dynamical systems are equivalent:

$$\{\hat{\mathbb{C}}; f(z) = \frac{1}{2} \left(z - \frac{1}{z} \right)\} \text{ and } \{\hat{\mathbb{C}}; f(z) = z^2\}.$$

4 A Rich Source for Fractals: Invariant Sets of Continuous Open Mappings

Let f be a transformation that acts on a space $\mathbf{X}$. Recall that a set A is invariant under f if $f^{-1}(A) = A$, and that this implies $f(A) = A$. We are interested in invariant sets of f that belong to $\mathcal{H}(\mathbf{X})$. The following theorem is a theoretical tool that provides both the existence and a means for the computation of invariant sets. Most of the material in this chapter is based on it.

Theorem 4.1 *Let (Y, d) be a metric space. Let $\mathbf{X} \subset \mathbf{Y}$ be compact and nonempty. Let $f : \mathbf{X} \rightarrow \mathbf{Y}$ be continuous and such that $f(\mathbf{X}) \supset \mathbf{X}$. Then, (1) a transformation $W : \mathcal{H}(\mathbf{X}) \rightarrow \mathcal{H}(\mathbf{X})$ is defined by*

$$W(A) = f^{-1}(A) \quad \text{for all } A \in \mathcal{H}(X).$$

(2) W possesses a fixed point $A \in \mathcal{H}(X)$, given by

$$A = \bigcap_{n=0}^{\infty} f^{\circ(-n)}(X) = \lim_{n \rightarrow \infty} W^{\circ n}(X).$$

Suppose f obeys the additional condition that $f(\mathcal{O})$ is an open subset of the metric space $(f(X), d)$ whenever $\mathcal{O} \subset X$ is an open subset of the metric space (X, d) . Then

(3) W is a continuous transformation from the metric space $(\mathcal{H}(X), h(d))$ into itself.

Proof of (1) and (2). The proof of (3) can be found in (Barnsley 1988c). (1) We begin by proving that W maps $\mathcal{H}(X)$ into $\mathcal{H}(X)$. Let $B \in \mathcal{H}(X)$. The condition $f(X) \supset X$ implies that $f^{-1}(B) \subset X$, and that $f^{-1}(B)$ is nonempty. B is compact, so it is a closed set in the metric space (X, d) . It follows that $X \setminus B$ is open. The continuity of f implies that $f^{-1}(X \setminus B)$ is open. Since $f(X) \supset X \supset B$, it follows that $f^{-1}(B) = X \setminus f^{-1}(X \setminus B)$. Hence $f^{-1}(B)$ is closed in the metric space (X, d) . Since X is compact it follows that $f^{-1}(B)$ is compact. This completes the proof of (1).

(2) Since $f(X) \supset X$, it follows that $X \supset f^{\circ(-1)}(X)$. Application of $f^{\circ(-n)}$ to both sides of the latter equation yields

$$X \supset f^{\circ(-1)}(X) \supset f^{\circ(-2)}(X) \supset f^{\circ(-3)}(X) \dots \supset f^{\circ(-n)}(X) \supset \dots$$

It follows that $\{f^{\circ(-n)}(X)\}$ is a Cauchy sequence in $\mathcal{H}(X)$, and it possesses a limit $A \in \mathcal{H}(X)$, given by

$$A = \bigcap_{n=0}^{\infty} f^{\circ(-n)}(X) = \lim_{n \rightarrow \infty} W^{\circ n}(X).$$

It remains to be proved that A is a fixed point of f . We need to show that $f^{\circ(-1)}(\bigcap_{n=0}^{\infty} A_n) = \bigcap_{n=0}^{\infty} A_n$ where $A_n = f^{\circ(-n)}(X)$ for $n = 1, 2, \dots$. First we prove that $f^{\circ(-1)}(\bigcap_{n=0}^{\infty} A_n) \supset \bigcap_{n=0}^{\infty} A_n$. Suppose $x \in f^{\circ(-1)}(\bigcap_{n=0}^{\infty} A_n)$. Then there is $y \in \bigcap_{n=0}^{\infty} A_n$ such that $x = f^{\circ(-1)}(y)$, and $y \in A_n$ for $n = 0, 1, 2, \dots$. It follows that $x \in f^{\circ(-1)}(A_n) = A_{n+1}$ for $n = 0, 1, 2, \dots$. It follows that $x \in \bigcap_{n=0}^{\infty} A_n$. To prove the inclusion the other way around, suppose that $x \in \bigcap_{n=0}^{\infty} A_n$. Then $x \in A_{n+1} = f^{-1}(A_n)$ for $n = 0, 1, 2, 3, \dots$. It follows that there is $y_n \in A_n$ such that $f(y_n) = x$ for $n = 0, 1, 2, 3, \dots$. The sequence $\{y_n\}$ possesses a convergent subsequence. Let y denote the limit of this subsequence. Then $y \in A_n$ for $n = 0, 1, 2, \dots$, and so $y \in \bigcap_{n=0}^{\infty} A_n$. Since f is continuous it follows that $f(y) = x$. Hence $x \in f^{\circ(-1)}(\bigcap_{n=0}^{\infty} A_n)$. We have shown that $f^{\circ(-1)}(\bigcap_{n=0}^{\infty} A_n) \subset \bigcap_{n=0}^{\infty} A_n$. This completes the proof of (2).

The invariant set A referred to in Theorem 4.1 can be expressed

$$A = \{x \in X : f^{\circ n}(x) \in X \text{ for all } n = 1, 2, 3, \dots\}.$$

That is, A is the set of points whose orbits do not escape from X . Also, A is the complement of the set of points whose orbits do escape. If $X \subset \mathbb{R}^2$, then pictures of A can be computed by using the Escape Time Algorithm.

The last statement in the theorem expresses a desirable property for a transformation W on $\mathcal{H}(\mathbf{X})$. If W is not continuous, yet $A_0 \in \mathcal{H}(\mathbf{X})$ and $\{W^n(A_0)\}$ converges to $A \in \mathcal{H}(\mathbf{X})$, one cannot conclude that $W(A) = A$. Without continuity of W , one should not trust the results of applying the Escape Time Algorithm. For example, slight numerical errors may mean that a computed sequence of sets $\{\tilde{A}_n \approx W^n(\mathbf{X})\}$ is not decreasing. One may still wish to define $\tilde{A} = \lim \tilde{A}_n$. Without continuity one cannot suppose that $f^{-1}(\tilde{A}) = \lim W(\tilde{A}_n) = \tilde{A}$, even approximately.

Analytic transformations map open sets to open sets. Hence their inverses act continuously on the space $\mathcal{H}(\mathbf{X})$, where $\mathbf{X} \subset \hat{\mathbb{C}}$ is chosen appropriately. To help visualize this, look back at Figure III.52. If the Sierpinski triangle ABO is deformed or moved, its inverse image $POQ \cup \tilde{P}O\tilde{Q}$ will move continuously with it.

The Hausdorff metric is the metric of perception: what we call a small change in the appearance of a picture is probably a small change in a Hausdorff distance. When one talks about continuous motion in the context of graphics, continuous growth in the context of botany, or continuous change in the context of a chemical system, the word “continuous” can often be replaced, pedantically, by “continuous in the Hausdorff metric.” Theorem 4.1 suggests that one could use continuous *open* maps to model such systems.

Examples & Exercises

***4.1.** Let $\lambda \in [-1, 1]$. Define a transformation $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$f(x, y) = \begin{cases} (x^2 - y^2 - 1, 2xy) & \text{when } x > 0, \\ (x^2 - y^2 - 1 + \lambda x, 2xy) & \text{when } x \leq 0. \end{cases}$$

Show that f is continuous. Show that if $\mathbf{X}$ denotes a ball, centered at the origin, of sufficiently large radius, then $f(\mathbf{X}) \supset \mathbf{X}$. Also verify that if $\lambda \in [-1, 0]$ then f maps open sets into open sets. Show that this is not the case for $\lambda = 1$. (Hint: look at what the map does to a very small disk centered at the origin.)

Figure VII.213 shows the result of applying the Escape Time Algorithm to f when $\lambda = 1$. The inner region, bounded by an ellipse, actually represents a disk $\mathbf{X}$ such that $f(\mathbf{X}) \supset \mathbf{X}$. Different scales have been used in the x and y directions. $f(\mathbf{X})$ is the region bounded by the outer curve. The image of a point that goes once around the inner ellipse is a point that goes twice around the origin, following the outer curve, which looks like a folded figure eight. Different “escape times” of orbits of points in $\mathbf{X}$ are represented by different gray tones. A magnified version of $\mathbf{X}$, painted by escape times, is shown in Figure VII.214. Roughly speaking, regions closest to the outside escape fastest. Points in the white region also escape. So where is the invariant set A ? It is right in the middle. It appears to be a branching, connected, tree-like set, with no interior.

Figure VII.215 shows the result of applying the Escape Time Algorithm to f when $\lambda = 0$. This time we see that the invariant set A , in the center, in white, is just the filled Julia set for $z^2 - 1$.

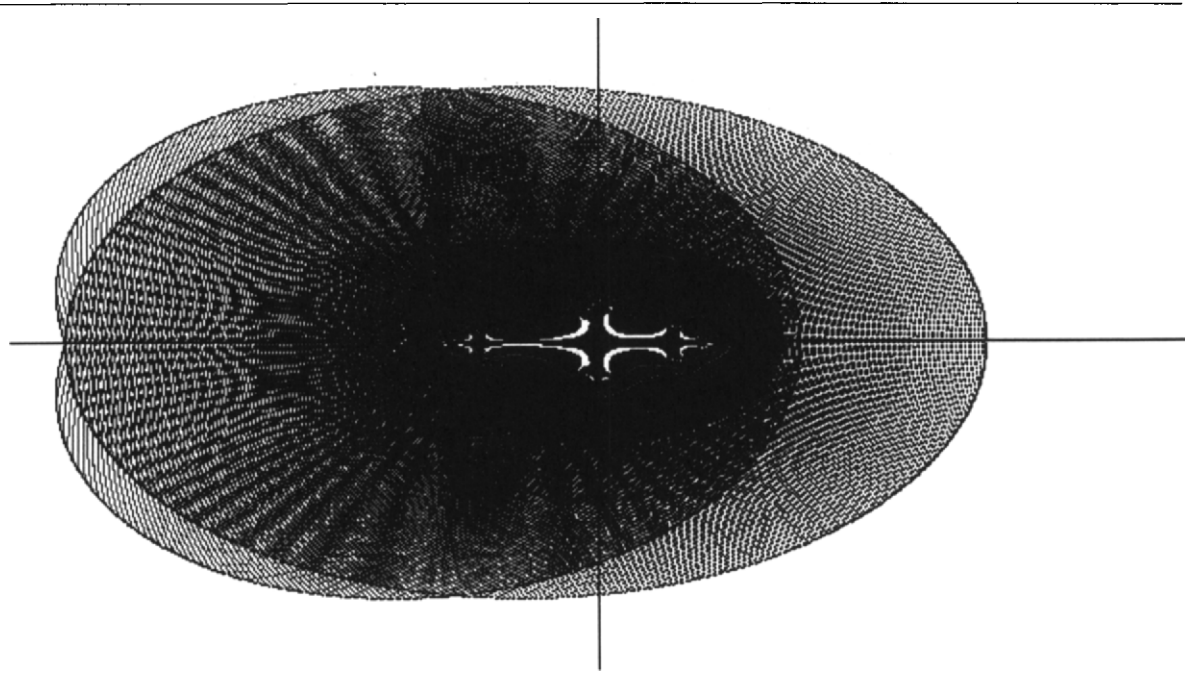


Figure VII.213. The result of applying the Escape Time Algorithm to the function f in example 4.1 with $\lambda = 1$. This function is continuous and is such that $f(\mathbf{X}) \supset \mathbf{X}$, where $\mathbf{X}$ denotes the Fabergé egg in the middle. $f(\mathbf{X})$ is the region bounded by the outer curve. Different “escape times” of orbits of points in $\mathbf{X}$ are represented by different gray tones.

What happens if we choose $\lambda = -1$? This time we obtain the image shown in Figure VII.216. However, this time things may not be as simple as they appear to be. The inner “layers” that surround the apparent invariant set A are highly irregular and unstable. That is, points that are very close together seem to have orbits that have very different escape times. Could it be that, although the mapping $W : \mathcal{H}(\mathbf{X}) \rightarrow \mathcal{H}(\mathbf{X})$ is continuous, it has a very poor modulus of continuity?

4.2. Construct a function $f : \mathbf{X} \rightarrow \mathbb{R}^2$, where $\mathbf{X} \subset \mathbb{R}^2$, which obeys the conditions of Theorem 4.1. Use the Escape Time Algorithm to analyze the associated invariant set A described in the statement of the theorem. Your example should be interesting and of a different character from those specifically described in this chapter.

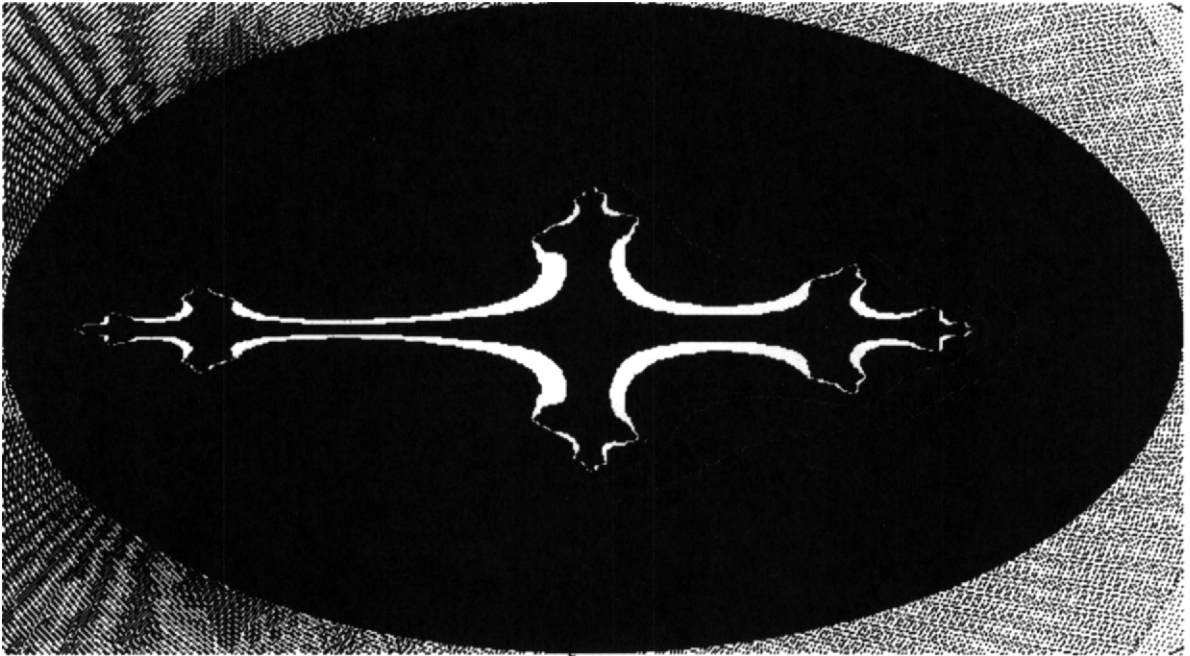


Figure VII.214. A magnified version of the region **X** in Figure VII.213. Approximately, orbits of points closest to the outside escape the fastest. Points in the white region also escape. The invariant set **A** is in the middle. It appears to be a branching, connected, tree-like set. It is surrounded by layers, just as the center of a real tree is surrounded by layers of growth.

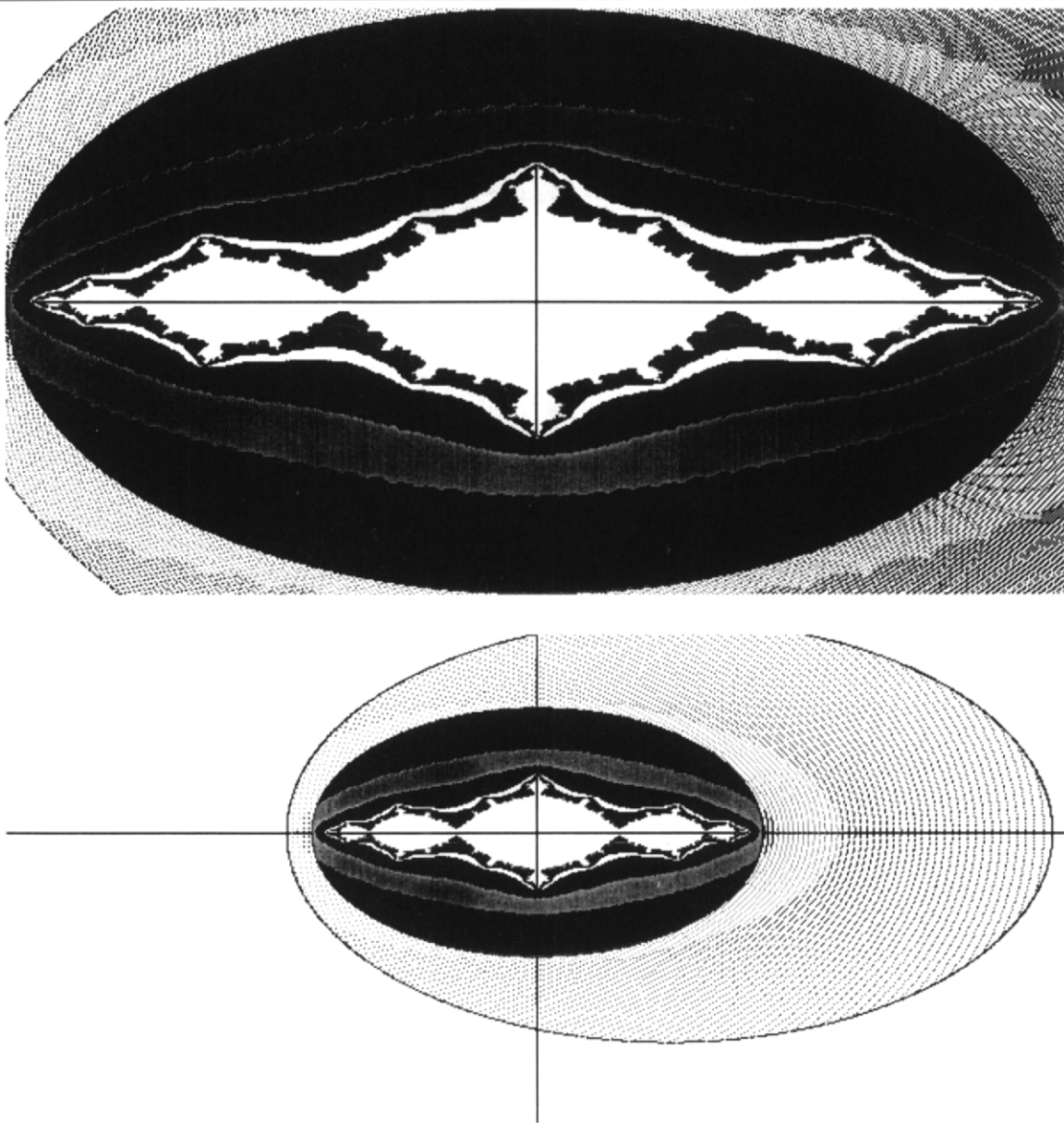


Figure VII.215. The result of applying the Escape Time Algorithm to f in example VII.213 with $\lambda = 0$. This time we see that the invariant set A , in the center, in white, is just the filled Julia set for $z^2 - 1$.

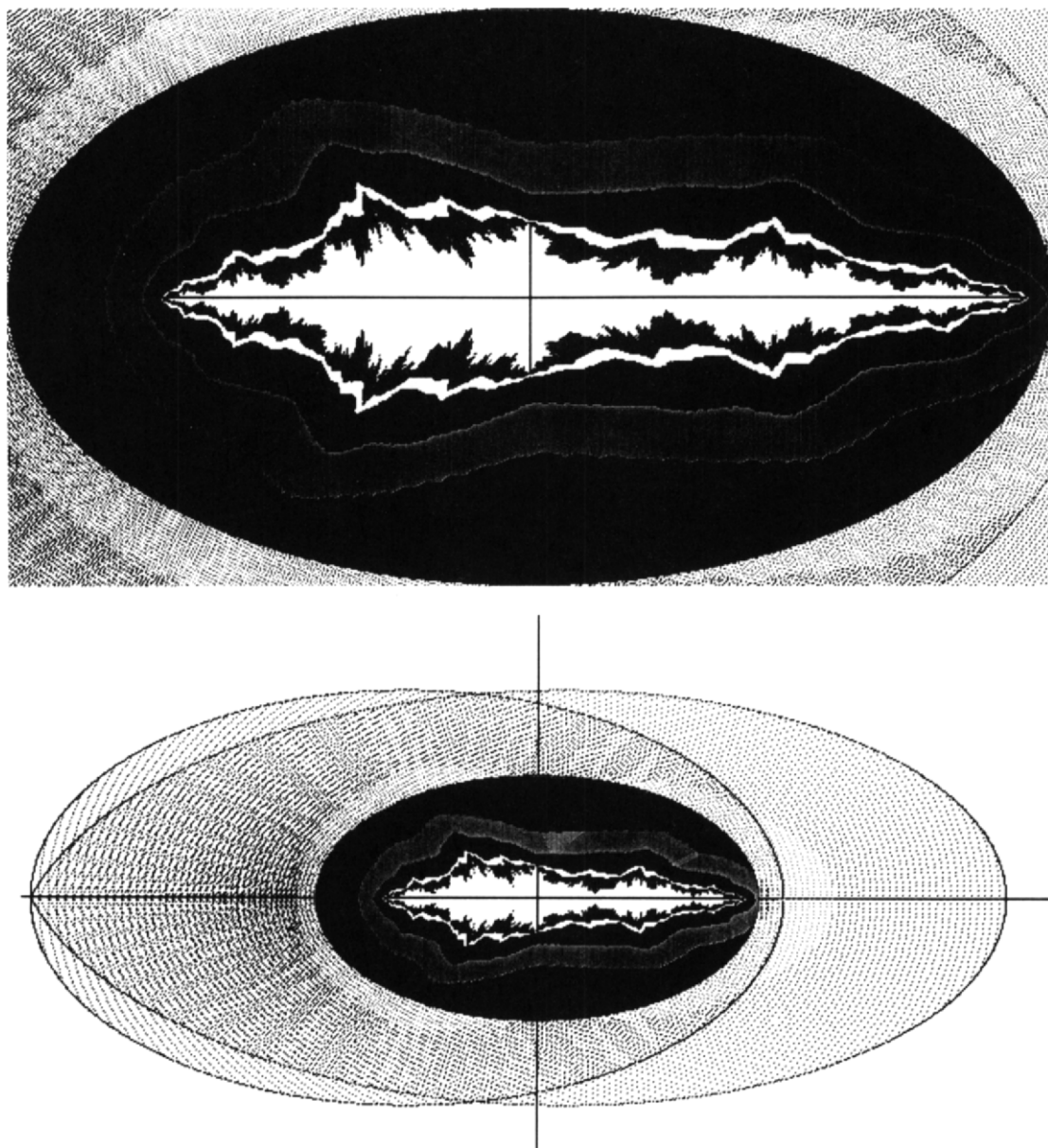


Figure VII.216. This image was computed using the Escape Time Algorithm applied to the Newton transformation associated with $f(z) = e^z - 1$. The viewing window is $\mathcal{W} = \{(x, y) \in \mathbb{R}^2 : -2.5 \leq x \leq 2.5, -2.5 \leq y \leq 2.5\}$. Can you work out what “escape region” was used?